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DASP: Drift-Attributed Subspace Plasticity for Localized Continual Adaptation in Streaming Network Intrusion Detection

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ABSTRACT Continual network intrusion detection must adapt to evolving traffic while preserving benign and previously learned attack behavior. Existing approaches commonly respond to drift by replaying a large buffer or updating the whole predictor, which couples unrelated feature groups, increases adaptation cost, and provides limited evidence about what changed. This paper presents Drift-Attributed Subspace Plasticity (DASP), a modular continual-learning framework that localizes distribution shift to feature-group subspaces, updates only the routed representation modules, and retains a shared trainable classifier. DASP combines layer- and module-wise discrepancy monitoring, a top-k attribution router, strict structural freezing, class-balanced bounded rehearsal, new-sample weighting, and a performance-gated emergency repair path. The method is evaluated under a prediction-first, one-window delayed-label protocol on five network-flow datasets, ten drift conditions, five stream seeds, two model seeds, and eight methods. The confirmation matrix contains 4,000 runs and compares DASP with seven locked alternatives. Across all eight methods, DASP obtains the highest observed mean Macro-F1 (0.9269), followed by SSF (0.9191) and Experience Replay (0.9036). SSF is the prespecified primary comparator: DASP wins 383 of 500 paired cells with a mean gap of $+0.0078$, and the locked-matrix sign test gives $p = 2.56 \times 10^{-34}$. DASP reduces adaptation time by a factor of 1.90 and gradient evaluations by a factor of 1.56 relative to SSF. The method performs no logged full-model or all-representation-module update; the shared classifier remains trainable, and all 16,000 observed repair records activate at most three of four representation modules. The main systems trade-off is lower forward throughput caused by the parallel modular backbone. These results support the integrated localized-repair system as an effective and auditable alternative to indiscriminate online retraining under the implemented supervision paths; they do not isolate the predictive contribution of the attribution router or establish a strict shared total-label-exposure comparison.

INDEX TERMS Concept drift, continual learning, delayed labels, intrusion detection, modular neural networks, parameter-efficient adaptation, replay, streaming evaluation.

I. INTRODUCTION

NETWORK intrusion detection systems (NIDSs) operate under non-stationary traffic, changing services, shifting attack prevalence, and evolving attacker behavior. A detector that is accurate on a static split may therefore become unreliable when the joint distribution $P_t(X, Y)$ changes over time. Concept drift research distinguishes abrupt, gradual, recurring, and local changes because they create different adaptation requirements [1]–[3]. In network-flow data, local class-conditional changes are particularly important: an attack family can change while benign traffic and other attack families remain comparatively stable.

The experimental protocol is part of the scientific problem. Randomly shuffling a stream removes the temporal structure that defines drift [4]. Training on the current window before producing its predictions leaks future information, while treating windows as independent replications exaggerates statistical evidence. Delayed labels make the setting more realistic but create a gap between prediction, querying, memory update, and supervised repair [5]. Severe class imbalance further makes accuracy an inadequate primary outcome [6], [7]; Macro-F1, attack recall, false-negative rate, and class-wise diagnostics are required.

Most continual NIDS methods use one of two strategies.

Replay methods mix current data with historical examples [8], [9]; protected-gradient methods constrain updates using a reference memory [10]; regularization methods preserve important parameters [8], [11]; and recent NIDS systems use strategic selection or representation protection [12], [13]. These mechanisms are valuable, but they generally answer the question "how much data should be used for adaptation?" rather than "which representation subspace is responsible for the drift?" Updating the full predictor can therefore alter stable features unnecessarily.

This paper introduces DASP, which changes the adaptation unit from the whole network to an attributed feature-group subspace. Its modular backbone partitions input features into parallel modules. A drift router computes discrepancy and supervised conflict signals for each module, selects at most k modules, and freezes every unselected module at the parameter level. A bounded class-balanced replay batch prevents the routed repair from becoming a short-memory classifier, while a delayed-label performance holdout selects emergency or background repair intensity. The resulting design is related to recent parameter-efficient adaptation [14]–[16] and modular continual learning [8], but is developed for delayed-label streaming NIDS evaluation.

The contributions are:

- a drift-attributed modular architecture that performs strict top-k subspace repair instead of global fine-tuning;
- a bounded localized rehearsal mechanism that combines delayed current labels, class-balanced memory, and new-sample weighting without changing the structural locality constraint;
- a reproducible ten-scenario evaluation with 1,200-run full pilot and 4,000-run confirmation matrices, eight locked methods, five datasets, five stream seeds, and two model seeds;
- a joint accuracy–adaptation–systems analysis that reports Macro-F1, attack recall, update time, gradient evaluations, throughput, memory, and localization invariants; and
- an explicit failure analysis that distinguishes mean superiority from worst-case cells and avoids claiming that every drift episode is won by DASP.

Accordingly, this study evaluates DASP through three research questions:

RQ1 How does DASP's stream-level predictive performance compare with seven locked online and continual-learning baselines across five datasets and ten controlled drift conditions?

RQ2 What structural-locality and adaptation-side efficiency properties does DASP exhibit relative to SSF under the same prediction-first delayed-label contract?

RQ3 Under which dataset, controlled-drift, and systems conditions do DASP's predictive or adaptation-cost advantages weaken or reverse?

These questions are comparative and evaluative rather than causal. The confirmation evaluates DASP as an integrated adaptation system; because random, fixed, and single-signal routing policies are not included in the locked matrix, the isolated contribution of attribution-guided routing to predictive performance is not identified.

II. RELATED WORK

A. CONCEPT DRIFT DETECTION AND STREAM EVALUATION

Recent reviews organize drift detectors into error-rate, adaptive-window, and distributional families [1], [4]. Representative foundations include DDM and ADWIN [17], [18], while the maximum mean discrepancy (MMD) compares samples in a reproducing-kernel Hilbert space without assuming a parametric form [19]. DASP uses MMD-like discrepancy signals as attribution evidence rather than as a standalone decision to retrain the entire network. The router is not a separately trained predictor: it is a deterministic scoring-and-selection component whose thresholds and top-k budget are locked before confirmation.

The distinction between detection and adaptation is important. A detector can identify a distribution shift without proving that the decision boundary changed. Conversely, performance degradation can appear before a distributional statistic crosses a threshold. DASP therefore combines representation discrepancy with delayed supervised conflict and a delayed-label performance holdout. This design follows the broader recommendation that drift type and drift locality should guide adaptation [2], [3].

B. CONTINUAL LEARNING AND SELECTIVE PLASTICITY

Experience replay remains a simple and effective continual-learning mechanism [8], [9]. A-GEM protects a new gradient using reference memory [10], while EWC constrains parameter changes using importance estimates [11]. Recent NIDS methods extend these ideas through strategic selection and representation protection [12], [13]. These approaches motivate DASP's rehearsal and structural protection, but DASP differs in making the protected unit an attributed feature module inside one predictor.

Parameter-efficient adaptation provides a complementary route. Adapter fusion and low-rank adaptation reduce the number of trainable parameters while retaining a fixed base model [14]–[16]. DASP applies the same efficiency principle to tabular network flows, but adds a data-driven routing signal and a strict inactive-module invariant. The router similarly asks where representations differ, then uses that difference to select plasticity. The routing rule is data-driven rather than learned end-to-end; its scores are computed from calibrated discrepancy and delayed-label conflict signals. This distinction matters because the confirmation experiment evaluates a fixed routing procedure, not a jointly optimized router.

C. CONTINUAL NIDS AND BENCHMARK DATA

Recent NIDS surveys identify UNSW-NB15 and CIC-IDS2017 as widely used benchmark families [20]–[22]. The ToN-IoT dataset extends the setting to telemetry-rich IoT

environments [23]. NetFlow-oriented datasets improve scalability and reduce dependence on payload inspection [24]–[26]. Recent continual NIDS methods such as augmented replay, SSF, and RepShield establish relevant comparison points [9], [12], [13]. DASP uses these families as locked comparators rather than selectively reporting only favorable methods.

D. INTERPRETABILITY AND STATISTICAL REPORTING

Interpretability methods such as LIME and SHAP explain predictions after the model has acted [27]–[29]. DASP exposes a structural explanation during adaptation: the router reports the selected modules, their plasticity values, and the replay budget. This is a different, intervention-oriented form of explanation. The paper reports paired stream-level comparisons, avoids window-level pseudo-replication, and uses exact paired tests together with effect sizes, consistent with recent guidance on machine-learning metrics and statistical tests [30].

E. SYNTHESIS AND RESEARCH GAP

Within the cited concept-drift, continual-learning, and continual-NIDS literature, three linked gaps remain. First, drift detectors usually answer whether a stream has changed, whereas a repair policy must also decide where parameter updates are admissible. Second, continual-learning methods protect prior knowledge mainly through replay, regularization, distillation, or gradient constraints applied to a globally shared predictor; these mechanisms do not by themselves provide a falsifiable module-level locality invariant. Third, continual NIDS studies often report predictive quality without jointly auditing delayed-label semantics, update cost, inference throughput, and the exact unit of statistical replication. DASP addresses this intersection by treating attribution as a routing decision, locality as a hard optimizer constraint, and stream evaluation as a paired deployment contract. This framing is narrower than a claim of universally better continual learning: the contribution is a testable combination of attribution, constrained repair, and auditability for streaming NIDS.

III. PROBLEM FORMULATION

Let a stream be a sequence of windows $\mathcal{W} = \{W_t\}_{t=1}^{T}$, where $W_t = \{(x_i, y_i)\}_{i=1}^{n_t}$, $x_i \in \mathbb{R}^d$, and $y_i \in \{0, 1\}$. At prediction time, labels in the current window are unavailable. After a fixed delay $\delta = 1$, delayed labels become available under the implemented method-specific supervision path. Primary DASP splits the labeled window into repair and holdout partitions and selects at most $B_{\text{train}} = 128$ repair examples. Hence, B_{train} bounds the optimization subset, not total label exposure. The detector must produce $\hat{y}_t$ before using any labels from W_t.

The task is deliberately scoped to binary benign-versus-attack prediction with one-window delayed supervision and bounded repair training. In this paper, *attribution* means an operational assignment of repair responsibility from discrepancy and delayed-label conflict scores; it does not denote causal feature attribution. The classifier is a shared adaptation group, while the locality claim concerns the four representation modules. The goal is not only to maximize predictive quality, but to minimize unnecessary parameter motion. Let Θ denote all parameters and $\mathcal{A}_t \subseteq \{1, \ldots, M\}$ denote the attributed active module set. DASP seeks

$$\max \ \mathbb{E}[\text{MacroF1}] - \lambda_u \,\text{UpdateCost} - \lambda_m \,\text{Motion}(\Theta_{\neg \mathcal{A}_t}), \tag{1}$$

subject to $|\mathcal{A}_t| \le k$, where $k = 3$ in the locked confirmation configuration. The last term encodes the structural preservation requirement: inactive modules must not change during a repair.

IV. DRIFT-ATTRIBUTED SUBSPACE PLASTICITY

A. DESIGN RATIONALE AND SYSTEM STATE

A conventional depth-wise MLP exposes globally shared hidden layers, so an update motivated by one feature group can move parameters used by every group. DASP instead creates four parallel representation modules and a shared fusion classifier. The modules are not assumed to be causally independent or intrinsically human-interpretable. They are a controlled intervention surface: each parameter group can be excluded from the optimizer and verified independently, while the classifier remains trainable because concept drift may move the decision boundary even when representation discrepancy is modest. Replay supplies historical supervision, whereas the routing mask determines where that supervision may alter parameters.

The persistent state at iteration t is

$$\mathcal{S}_t = \left(\Theta_t, \mathcal{M}_t, \mathcal{H}_t^{\text{ref}}, \mathcal{Q}_t\right), \tag{2}$$

where Θ_t is the model state, $\mathcal{M}_t$ is bounded replay memory, $\mathcal{H}_t^{\text{ref}}$ is the reference-representation bank, and $\mathcal{Q}_t$ is the pending-update queue. Before labels for W_t are available, the online procedure computes representation evidence $\mathcal{E}_t = (\mathbf{r}_t, s_t, D_t)$, emits $\hat{Y}_t$, and enqueues $(W_t, \hat{Y}_t, \mathcal{E}_t)$. Let $\tau = t - 1$. When Y_τ arrives, the procedure retrieves the stored item for W_τ, restores $\mathcal{E}_\tau$, computes $\mathbf{c}_\tau$, $\mathbf{p}_\tau$, and $\mathcal{A}_\tau$, and updates the model for subsequent windows. Thus,

$$\mathcal{S}_{t+1} = \mathcal{T}(\mathcal{S}_t, W_t, \mathcal{E}_t, Y_\tau, \mathcal{Q}_t[\tau]), \qquad \tau = t - 1, \tag{3}$$

and no routing quantity indexed by t is combined with labels from W_{t-1}.

B. MODULAR FEATURE-SUBSPACE BACKBONE

The backbone partitions the standardized input into $M = 4$ contiguous groups. Module j maps its feature slice $x^{(j)}$ into a 64-dimensional representation,

$$h_j = f_j(x^{(j)}; \theta_j), \qquad j = 1, \ldots, M, \tag{4}$$

and the shared classifier produces

$$z = g([h_1 \| h_2 \| \cdots \| h_M]; \phi), \qquad \hat{y} = \arg\max_c z_c. \tag{5}$$

The same fixed partition is used for every stream cell. Its purpose is parameter-level controllability, not a claim that contiguous groups correspond to independent causal mechanisms.

C. WARM-UP CALIBRATION AND MMD REQUIREMENT

Warm-up trains the initial predictor and fills $\mathcal{M}_t$. The implementation then freezes a reference-model snapshot and constructs $\mathcal{H}^{\text{ref}}$ from the final warm-up window. The four calibration representation sets are produced by this same frozen reference model on the last four warm-up windows. For each module, calibration compares the fixed final-window bank with each calibration-window representation set; it is neither leave-one-window-out nor based on an independently sampled reference. For module j, DASP subsamples at most $q = 128$ reference and current representations, computes squared pairwise distances over the joined sample, and uses the median positive distance $b_{j,t}$ as the base bandwidth. The multi-scale kernel is

$$K_{j,t}(u,v) = \frac{1}{3}\sum_{s\in\{0.5,1,2\}} \exp\left(-\frac{\|u-v\|_2^2}{2b_{j,t}s^2}\right). \tag{6}$$

The implementation uses the non-negative biased estimate

$$m_{j,t} = \left[\frac{1}{n^2}\sum_{a,b}K(u_a,u_b) + \frac{1}{m^2}\sum_{a,b}K(v_a,v_b) - \frac{2}{nm}\sum_{a,b}K(u_a,v_b)\right]_+ \tag{7}$$

Across the four calibration windows, let $\widetilde{\mu}_j$ and MAD_j denote the median and median absolute deviation of $m_{j,t}$. DASP computes

$$z_{j,t} = \max\left(0, \frac{m_{j,t}-\widetilde{\mu}_j}{1.4826\max(\mathrm{MAD}_j, 10^{-6})+\epsilon}\right), \tag{8}$$

$$r_{j,t} = \mathrm{clip}\left(\frac{z_{j,t}-\kappa}{z_{\max}-\kappa}, 0, 1\right), \tag{9}$$

with $\kappa = 2$, $z_{\max} = 12$. Severity is the mean of the largest $\min(k, M)$ requirement scores. In the evaluated DASP configuration, a drift episode is triggered when severity exceeds $\theta_s = 0.15$ for $P = 1$ consecutive window (equivalently, the current window alone). The locked experimental configuration also specifies an emergency-severity value of 0.60, but this parameter is not used by the evaluated primary procedure and is therefore inactive in the reported DASP results.

D. DELAYED CONFLICT AND TOP-K ROUTING

Let $\tau = t-1$ denote the delayed window being repaired at outer iteration t. The implementation partitions labeled W_τ into a 75% repair pool and a disjoint 25% performance holdout, then uses repair-pool labels to select at most 128 class-balanced training examples. Thus, 128 is a training-subset count; the matrix compares the implemented supervision paths rather than a shared 128-total-label-query protocol.

Let C_τ^{new} be the sampled delayed repair-training batch and B_τ^{mem} a class-balanced replay batch. DASP compares current-window and replay gradients for each adaptation module,

$$c_{j,\tau} = \max\left(0, -\frac{\langle g_{j,\tau}^{\text{new}}, g_{j,\tau}^{\text{mem}}\rangle}{\|g_{j,\tau}^{\text{new}}\|_2\|g_{j,\tau}^{\text{mem}}\|_2+\epsilon}\right). \tag{10}$$

A high $c_{j,\tau}$ indicates greater forgetting risk. During a triggered attributed repair, the stored requirement for the same delayed window is attenuated as

$$p_{j,\tau} = \frac{r_{j,\tau}}{1+\lambda_c c_{j,\tau}}, \qquad \lambda_c = 1. \tag{11}$$

The active set contains at most the $k = 3$ modules with the largest positive $p_{j,\tau}$; if none is positive, the maximum-scoring module is selected. The no-trigger branches rank modules by conflict and use a deterministic first-layer fallback if the active list is empty.

The delayed-label performance holdout is evaluated before optimization and only selects emergency versus background intensity. The evaluated primary DASP procedure applies the selected update directly; it has no post-update candidate acceptance or rollback step.

E. REPLAY, MEMORY LIFECYCLE, AND STRICT FREEZING

The localized rehearsal batch is

$$\widetilde{C}_\tau = C_\tau^{\text{new}} \cup B_\tau^{\text{mem}}, \tag{12}$$

where $|B_\tau^{\text{mem}}| \le 512$, current examples receive weight $w_{\text{new}} = 3$, and replay examples receive unit weight. The weighted loss is

$$\mathcal{L}_\tau = \frac{\sum_{(x_i,y_i)\in\widetilde{C}_\tau} w_i\,\ell(g(f(x_i)),y_i)}{\sum_i w_i}. \tag{13}$$

The memory capacity is 1,024. After localized training, C_τ^{new} is inserted into memory. Reference refresh is enabled in the locked confirmation. After every applied primary DASP repair—triggered, emergency, or background—the updated model is deep-copied as the new reference snapshot, reference representations are recomputed on W_τ, and calibration is recomputed over the recent windows ending at τ.

For selected modules, the learning rate is scaled by the mode-specific plasticity. The structural mask is

$$\eta_{j,\tau} = \begin{cases} p_{j,\tau}, & j \in \mathcal{A}_\tau, \\ 0, & j \notin \mathcal{A}_\tau, \end{cases} \qquad \eta_{\text{classifier},\tau} = 1. \tag{14}$$

Zero-plasticity modules are omitted from the AdamW optimizer rather than merely assigned a zero gradient. Consequently, neither gradient updates nor decoupled weight decay can change inactive modules. Pre- and post-update parameter-state fingerprints, together with the active-module count, are recorded to verify these structural invariants.

F. END-TO-END WINDOW ALGORITHM

Figure 1 presents the complete implemented DASP transition, while Fig. 2 provides the corresponding pseudocode. These descriptions specify the complete transition without assuming a factorial contribution from any individual component.

G. COMPLEXITY, WORKED EXAMPLE, AND INTERPRETABILITY

Let P_j be the parameters of module j, P_c the classifier parameters, q the MMD subsample size, h the module representation

TABLE 1. Implementation-faithful repair policy for the reported primary DASP path. The classifier multiplier is 1 on the base learning rate 10^{-3} in every mode.

Mode	Condition for delayed window τ	Routed representation modules	Module plasticity	Epochs
Triggered attributed repair	Stored MMD severity triggers at $\theta_s = 0.15$ with $P = 1$	Up to three positive top-k values of $p_{j,\tau}$; maximum-score fallback	$p_{j,\tau} = r_{j,\tau}/(1 + c_{j,\tau})$	2
Performance-gated emergency repair	No MMD trigger and $F_\tau^{\text{hold}} < 0.80$	Three highest-conflict modules; deterministic fallback if needed	$\min\{1, \max(0.25, c_{j,\tau})\}$	3
Background local consolidation	No MMD trigger and $F_\tau^{\text{hold}} \geq 0.80$	Three highest-conflict modules; deterministic fallback if needed	0.25	2

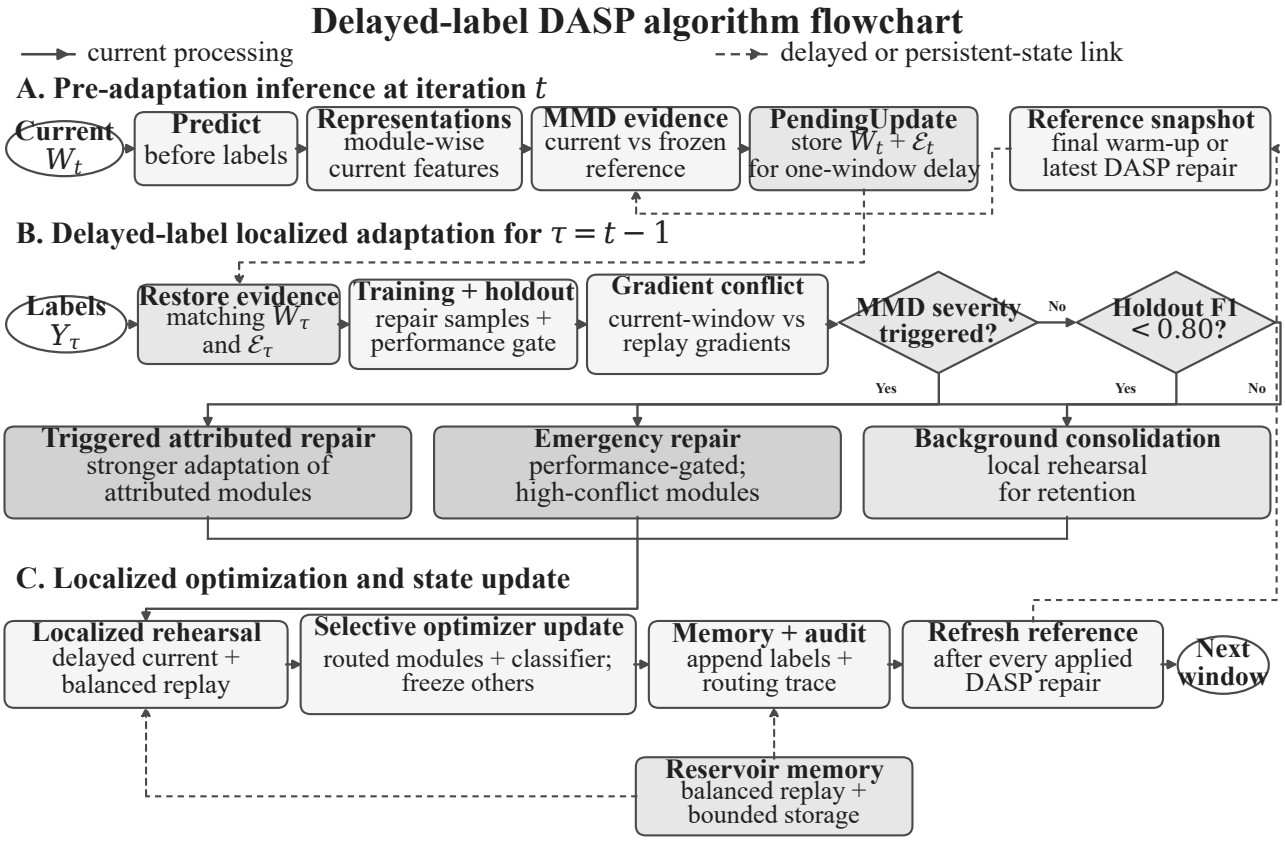


FIGURE 1. Overview of the one-window delayed-label DASP adaptation cycle. For each current window W_t, predictions and MMD evidence are computed before label access and stored in a pending-update queue. At the same outer iteration, labels for W_τ, $\tau = t - 1$, restore the matching evidence; module-wise gradient conflict and the delayed-label performance holdout then determine repair intensity. The selected localized update is applied directly, unaffected modules remain frozen, and memory, reference, and audit state are updated. The reported primary path does not implement post-update candidate rollback.

width, B the rehearsal batch size, and E_t the selected epoch budget. The biased MMD computation costs $O(Mq^2h)$, conflict estimation requires two gradient calculations over the adaptation groups, and localized repair costs $O(E_tB(P_c + \sum_{j\in\mathcal{A}_t} P_j))$. Replay and reference storage require $O(B_{\text{mem}}d + Mqh)$. All four modules are still evaluated at inference, explaining why update work can decrease while forward throughput remains lower than a flat MLP.

As an arithmetic illustration, let $\mathbf{r}_\tau = (0.18, 0.82, 0.46, 0.70)$ and $\mathbf{c}_\tau = (0.10, 0.22, 0.61, 0.18)$. Equation (11) gives $\mathbf{p}_\tau \approx (0.164, 0.672, 0.286, 0.593)$; with $k = 3$, modules $\{2, 4, 3\}$ and the classifier are updated, while module 1 remains byte-identical. This example illustrates the implemented calculation and is not presented as an empirical ablation result.

The routing record is therefore an operational intervention record: it stores MMD, robust scores, conflict, plasticity, selected modules, replay size, repair mode, gradient evaluations, and whether any full-model update occurred. It explains the scope of parameter change rather than providing post-hoc feature importance [27]–[29].

FIGURE 2. Prediction-first DASP pseudocode

Warm up f_Θ; initialize $\mathcal{M}$, $\mathcal{H}^{\text{ref}}$, and pending queue $\mathcal{Q}$.
for each outer iteration t with current window W_t **do**
  Predict $\hat{Y}_t = f_{\Theta_t}(X_t)$ before using labels from W_t.
  Compute and store $\mathcal{E}_t = (\mathbf{r}_t, s_t, D_t)$ from W_t and $\mathcal{H}^{\text{ref}}_t$.
  Enqueue $(W_t, \hat{Y}_t, \mathcal{E}_t)$ in $\mathcal{Q}$.
  if labels are available for $\tau = t - 1$ **then**
    Dequeue $(W_\tau, \hat{Y}_\tau, \mathcal{E}_\tau)$ and restore $\mathbf{r}_\tau, s_\tau, D_\tau$.
    From delayed labels, split repair pool/holdout and form class-balanced C^{new}_τ with at most 128 training examples; draw B^{mem}_τ.
    Compute module conflicts $c_{j,\tau}$ from C^{new}_τ and B^{mem}_τ gradients.
    if stored MMD decision D_τ triggers **then**
      Route positive top-k $p_{j,\tau} = r_{j,\tau}/(1 + c_{j,\tau})$; use maximum-score fallback.
    else if $F^{\text{hold}}_\tau < 0.80$ **then**
      Route top-k conflicts for three-epoch performance-gated emergency repair.
    else
      Route top-k conflicts for two-epoch background local consolidation with $p = 0.25$.
    end if
    Train on $\widetilde{C}_\tau = C^{\text{new}}_\tau \cup B^{\text{mem}}_\tau$.
    Update routed modules and classifier; exclude unaffected modules from AdamW.
    Update memory and audit; refresh the enabled reference state using W_τ.
    Commit Θ_{t+1} for subsequent windows; no candidate rollback is executed.
  end if
end for

V. EXPERIMENTAL DESIGN

A. DATASETS AND TASK

We evaluate binary benign-versus-attack detection on five network-flow collections: CIC-IDS2017, NF-BoT-IoT-v3, NF-CSE-CIC-IDS2018-v3, NF-ToN-IoT-v3, and NF-UNSW-NB15-v3. The classical source datasets are described in [20]–[23]; the NetFlow construction and dataset portal are documented in [24]–[26]. All methods use the same chronologically ordered experimental streams. No future window contributes to a prediction.

B. TEN PRIMARY DRIFT CONDITIONS

The protocol contains ten conditions: stationary control ($S0$), sudden drift ($S1$), gradual drift ($S2$), incremental drift ($S3$), recurrent drift ($S4$), transient blip ($S5$), combined drift ($S6$), and local class-conditional drift at low, medium, and high intensity ($S7$-Local-Low/Medium/High). Local scenarios change the composition associated with selected classes while retaining non-target traffic as an anchor. The taxonomy follows the distinction between drift type and locality in [1]–[3].

TABLE 2. Dataset and stream protocol.

Dataset	Role in evaluation
CIC-IDS2017	Enterprise traffic with multiple attack families
NF-BoT-IoT-v3	IoT botnet and denial-of-service behavior
NF-CSE-CIC-IDS2018-v3	Enterprise/IoT flow variant with evolving attack composition
NF-ToN-IoT-v3	Telemetry-rich IoT and network traffic
NF-UNSW-NB15-v3	Heterogeneous modern network-flow benchmark

C. METHODS AND FAIRNESS CONTROLS

The eight methods are locked before evaluation: Online MLP, Experience Replay, Online EWC, A-GEM, SSF, RepShield, CDDA-MD, and modular DASP. All methods receive the same prepared stream, one-window prediction-first delay, and seed assignment. Their supervision paths differ: primary DASP selects 128 class-balanced repair examples from a labeled pool and evaluates a disjoint 25% holdout; primary SSF also selects after label access, whereas the budget-matched comparators use label-blind uniform selection. The matrix therefore does not establish equal total label exposure. Method-specific controls and this exception are reported explicitly in Table 21 and Appendix C. The SSF implementation follows strategic sample selection and forgetting protection [12]; RepShield follows representation-protection motivation [13]; the remaining baselines represent direct online training, replay, regularization, and gradient projection.

D. PILOT AND CONFIRMATION MATRICES

The full pilot contains $5 \times 10 \times 3 \times 1 \times 8 = 1{,}200$ runs. The confirmation matrix contains $5 \times 10 \times 5 \times 2 \times 8 = 4{,}000$ runs, with stream seeds $\{2001, 2002, 2003, 2101, 2102\}$ and model seeds $\{91, 92\}$. Each dataset–scenario–seed–model cell is paired across methods. The confirmation has 500 DASP–SSF paired comparisons.

E. METRICS AND STATISTICAL ANALYSIS

The primary metric is Macro-F1 because it weights benign and attack classes equally. We also report attack recall, false-negative rate, cumulative Macro-F1, online forgetting, update time, median update time, gradient evaluations, prediction throughput, memory payload, and peak resident memory. Metric choice under class imbalance should combine class-balanced summaries with class-wise error analysis [7], [30].

The analysis unit is the paired dataset–scenario–stream-seed–model-seed cell, not the window. We report paired mean gaps, medians, wins, and a sign test. Cells are paired exactly across methods, but cells sharing a dataset or scenario generator are not claimed to be independent draws from all possible operational deployments. We additionally use a paired t-test and Wilcoxon signed-rank test as sensitivity analyses, with Holm correction when multiple exploratory comparisons are made [30]. Bootstrap intervals may be added after the

final archival run; the present analysis emphasizes the exact paired results to avoid treating highly correlated windows as independent. The very small nominal p-values are therefore interpreted as sensitivity evidence for the locked matrix, while the dataset-wise, scenario-wise, and worst-cell summaries carry the generalization argument.

F. REPRODUCIBILITY AND PERFORMANCE MEASUREMENT

All runs use single-thread CPU execution, fixed stream and model seeds, a locked experimental configuration, and deterministic run identifiers. Adaptation time is the sum of update intervals, excluding data preparation. Prediction throughput is measured separately from update time, so a faster predictor is not conflated with a faster adaptation algorithm. The same streaming evaluation procedure is applied consistently across the full protocol.

VI. RESULTS

A. OVERALL CONFIRMATION RESULTS

Table 3 is generated directly from the 4,000-run confirmation. DASP obtains the highest observed mean Macro-F1, 0.9269, followed by SSF at 0.9191 and Experience Replay at 0.9036. The DASP–SSF mean gap is $+0.0078$ and the median gap is $+0.0032$. DASP wins 383 of 500 paired cells and remains within ± 0.01 of SSF in 303 cells. The sign test gives $p = 2.56 \times 10^{-34}$; the paired t-test and Wilcoxon sensitivity tests give $p = 4.59 \times 10^{-20}$ and $p = 2.39 \times 10^{-27}$, respectively.

B. DATASET-LEVEL ANALYSIS

Table 4 summarizes all methods by dataset, while Fig. 3 shows the method distribution. DASP is the best method on CIC-IDS2017 (0.9420), NF-BoT-IoT-v3 (0.8652), NF-CSE-CIC-IDS2018-v3 (0.9627), and NF-ToN-IoT-v3 (0.8684). On NF-UNSW-NB15-v3, several methods are effectively saturated near 0.996; DASP reaches 0.9961, while A-GEM, Experience Replay, RepShield, and Online EWC reach 0.9962. Thus, the last dataset is a ceiling case rather than a decisive separation between methods.

The largest positive DASP–SSF dataset gap is on NF-BoT-IoT-v3, $+0.0215$. The difference is compatible with the design hypothesis that class-conditional botnet changes can often be repaired through a subset of feature modules rather than by updating all parameters; the confirmation matrix does not by itself establish that this mechanism is causal. NF-ToN-IoT-v3 remains the most difficult dataset, but DASP still exceeds SSF by $+0.0042$ on average. The large negative outlier is NF-BoT-IoT-v3, $S2$-Gradual, stream seed 2102, model seed 91, where DASP trails SSF by 0.0900; this cell is retained as a failure case rather than removed.

C. DRIFT-CONDITION ANALYSIS

Fig. 4 shows that the mean DASP–SSF gap is positive in all ten scenarios. The largest mean gains occur for S7-Local-Low ($+0.0141$), S0 ($+0.0131$), S7-Local-Medium ($+0.0110$), and S5-Blip ($+0.0110$). S7-Local-High is approximately tied ($+0.0002$), and S2-Gradual is also close ($+0.0014$). This pattern suggests that attribution is most useful when the drift is localized or transient, while high-intensity local drift can activate several modules and reduce the advantage of selectivity.

Figs. 5–9 provide the corresponding window-level view for all eight methods. The trajectories reinforce the aggregate result but do not show uniform window-by-window dominance. On CIC-IDS2017, DASP ranks first overall at 0.9420, exceeds SSF by 0.0070, and has the highest scenario mean in seven of ten conditions. On NF-BoT-IoT-v3, it again leads seven conditions and has the largest dataset-level advantage over SSF ($+0.0215$), although S2-Gradual and S7-Local-High remain boundary cases. On NF-CSE-CIC-IDS2018-v3, DASP leads overall by only 0.0027 over Experience Replay and exceeds SSF in all ten scenario means, but it ranks seventh in S5-Blip. NF-ToN-IoT-v3 is more mixed: DASP leads five conditions and exceeds the strongest baseline by only 0.0032. NF-UNSW-NB15-v3 is a ceiling case: DASP ranks sixth at 0.9961, only 0.0001 below the leading group, while still exceeding SSF in every scenario mean. These crossings and exceptions support a bounded comparative-effectiveness claim rather than universal superiority.

D. ADAPTATION COST AND SYSTEMS PERFORMANCE

DASP's mean update time is 1.415 s per stream, compared with 2.683 s for SSF. It uses 328.7 gradient evaluations versus 512 for SSF, giving a 1.90-fold update speedup and a 1.56-fold gradient reduction. The advantage is stable across datasets: the update speedup ranges from 1.77 on NF-BoT-IoT-v3 to 2.03 on NF-UNSW-NB15-v3.

The main cost is prediction throughput. DASP reaches approximately 159,219 flows/s, whereas the other MLP-based methods are near 455,000–465,000 flows/s. The modular backbone executes four parallel feature modules and a fusion classifier, so the throughput reduction is an architectural cost of attribution. This is acceptable when adaptation time and auditability are primary constraints, but it should not be hidden in a single ''efficiency'' number. Fig. 10 reports the accuracy–update frontier explicitly.

E. LOCALIZATION AND STRUCTURAL PRESERVATION

Across the 500 DASP confirmation runs, 16,000 window-level repair records were available. The active-module count has minimum 1, median 3, and maximum 3; no record opens all four modules. The replay size is exactly 512 in the logged repair records. The global-update count is zero. These invariants distinguish DASP from an implementation that merely labels a full-model update as selective.

F. DEVELOPMENT ABLATION AND FAILURE ANALYSIS

The small development pilot illustrates why the final mechanism was required. The original no-update path reached mean Macro-F1 0.7625 against SSF 0.9211. Classifier calibration improved it to 0.8706, the early semi-local prototype to 0.8725, and bounded localized rehearsal with new-sample weighting to approximately 0.9266 on the same 15 paired cells. The

TABLE 3. Confirmation results over 4,000 runs. Values are means across five stream seeds and two model seeds for each dataset–scenario cell.

Method	Macro-F1	Update (s)	Grad. eval.	Throughput (flows/s)
modular dasp	0.9269	1.415	328.7	159,219
ssf	0.9191	2.683	512.0	455,279
experience replay	0.9036	0.546	192.0	461,493
repshield	0.8881	0.946	128.0	464,789
agem	0.8821	0.495	128.0	464,700
online mlp	0.8751	0.222	64.0	464,101
online ewc	0.8751	0.396	64.0	454,905
cdda md	0.7973	0.038	2.7	458,585

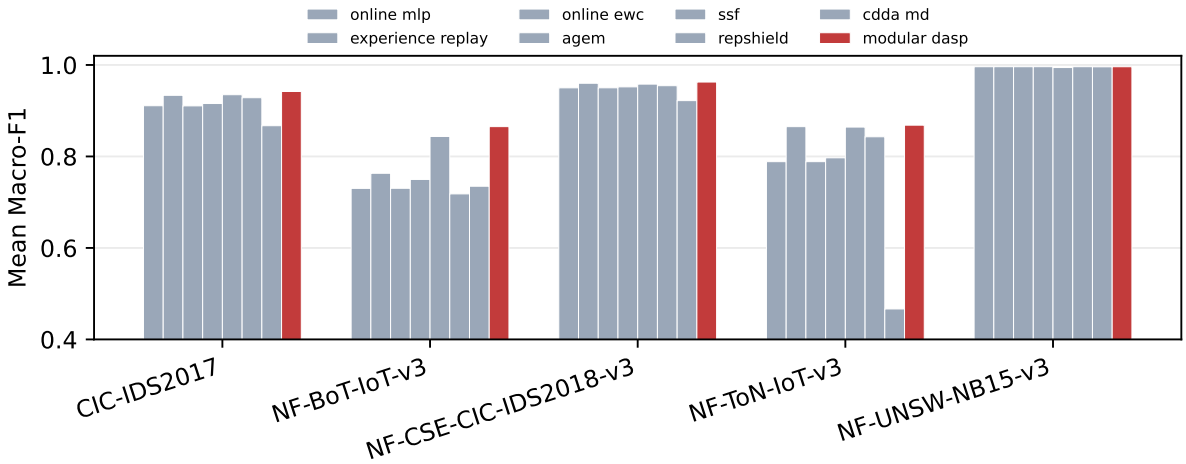


FIGURE 3. Mean Macro-F1 by dataset over the confirmation matrix. DASP is highlighted in red.

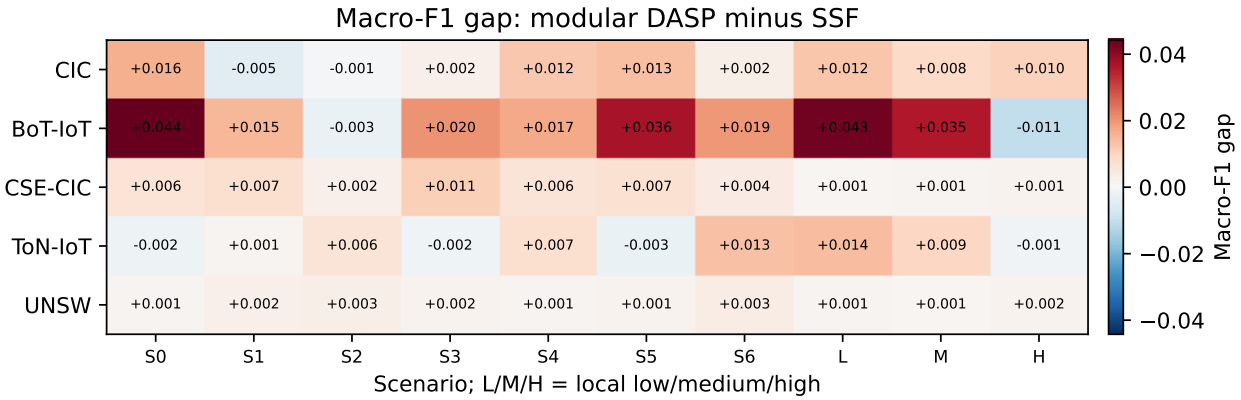


FIGURE 4. DASP minus SSF Macro-F1 gap by dataset and drift condition. Positive values are blue; negative values are red.

TABLE 4. Dataset-level mean Macro-F1 and update cost.

Dataset	Macro-F1	Update (s)
cic-ids2017	0.9179	0.808
nf-bot-iot-v3	0.7669	0.845
nf-cse-cic-ids2018-v3	0.9512	0.842
nf-ton-iot-v3	0.7852	0.860
nf-unsw-nb15-v3	0.9959	0.859

ablation is diagnostic, not a confirmation claim: it shows that the principal failure mode was under-adaptation, not an invalid locality hypothesis.

The worst confirmation cell is retained in the analysis because aggregate means can conceal instability. In the NF-BoT-IoT-v3 S2-Gradual cell with stream seed 2102 and model seed 91, DASP's Macro-F1 is 0.0900 below SSF. This failure is consistent with gradual attribution ambiguity: several modules can receive moderate evidence before the classifier has a stable boundary. Future work should test adaptive k, uncertainty-triggered expansion, and a boundary-focused performance

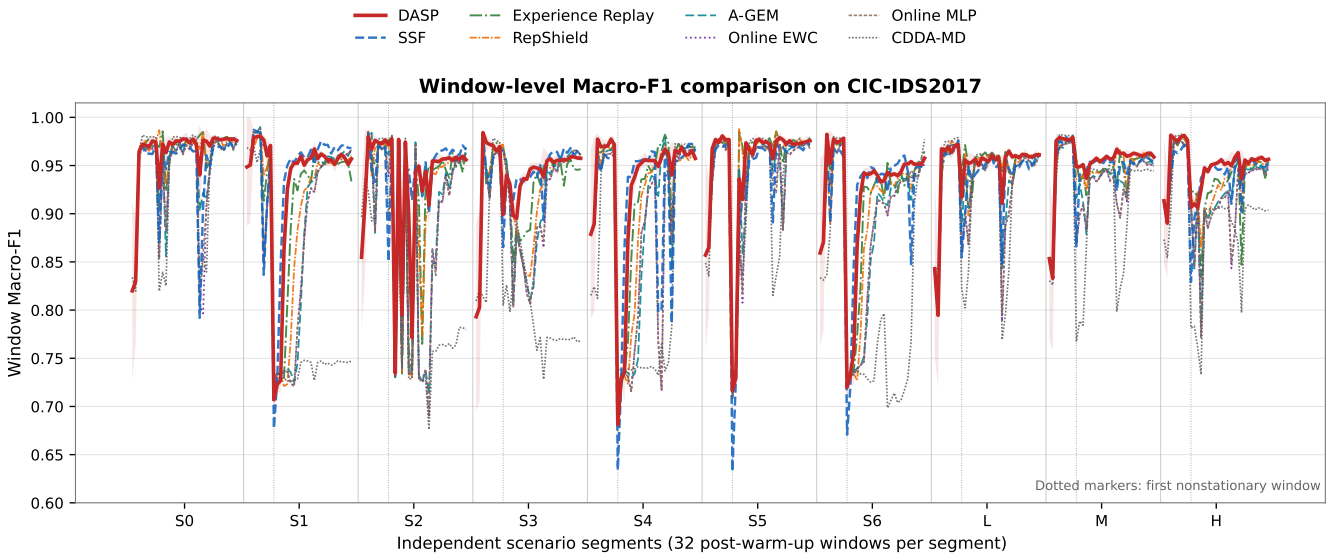


FIGURE 5. **Window-level Macro-F1 trajectories of all eight methods on CIC-IDS2017. Each line is the mean over five stream seeds and two model seeds; the red band is a 95% normal-approximation confidence interval for DASP. Scenario segments are independent runs, gaps prevent connections across resets, and dotted markers indicate the first nonstationary window. L/M/H denote low-, medium-, and high-intensity local drift.**

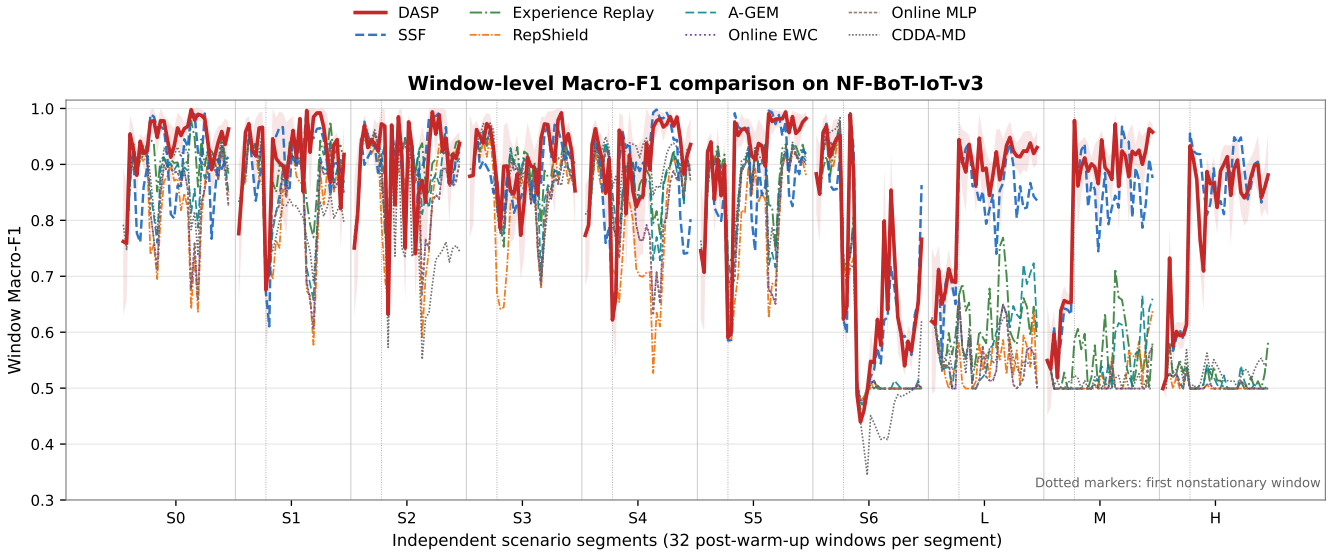


FIGURE 6. **Window-level Macro-F1 trajectories of all eight methods on NF-BoT-IoT-v3. Curves and uncertainty are aggregated over five stream seeds and two model seeds. DASP is emphasized in red; independent scenario segments are separated by gaps, and dotted markers identify the first nonstationary window.**

gate on this condition without changing the locked primary result.

VII. DISCUSSION

RQ1: Comparative predictive effectiveness

The answer to RQ1 is qualified rather than universal. DASP has the highest observed mean Macro-F1 among the eight locked methods, with 0.9269 compared with 0.9191 for SSF and 0.9036 for Experience Replay. Against the prespecified SSF comparator, the mean paired gap is $+0.0078$ and DASP wins 383 of 500 cells. The effect is therefore consistent across much of the locked matrix, but its magnitude is modest and the cell-level distribution contains meaningful losses. The evidence supports comparative effectiveness under the evaluated delayed-label protocol, not unconditional superiority.

RQ2: Structural locality and adaptation efficiency

For RQ2, the locked implementation exhibits clear structural-locality and adaptation-efficiency properties. No repair opens all four representation modules, the global-update counter remains zero, and inactive modules are excluded from the optimizer; the shared classifier is the declared exception to the representation-module invariant. Relative to SSF, DASP

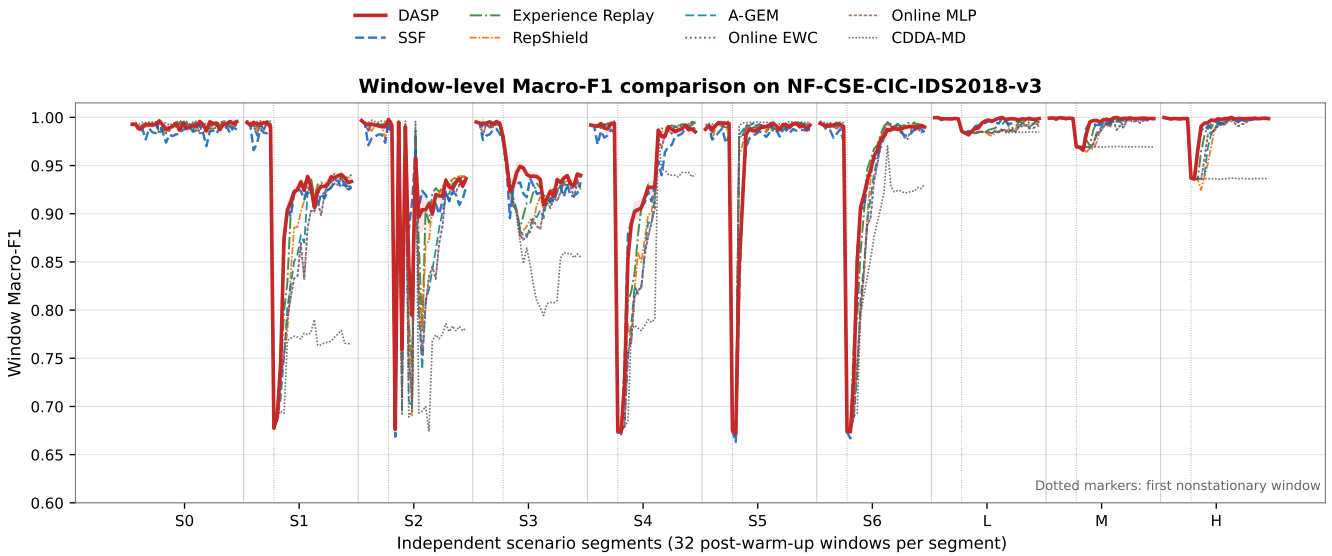


FIGURE 7. **Window-level Macro-F1 trajectories of all eight methods on NF-CSE-CIC-IDS2018-v3. The red DASP curve includes a 95% normal-approximation confidence band. Segment gaps denote independent scenario resets; dotted markers indicate drift onset.**

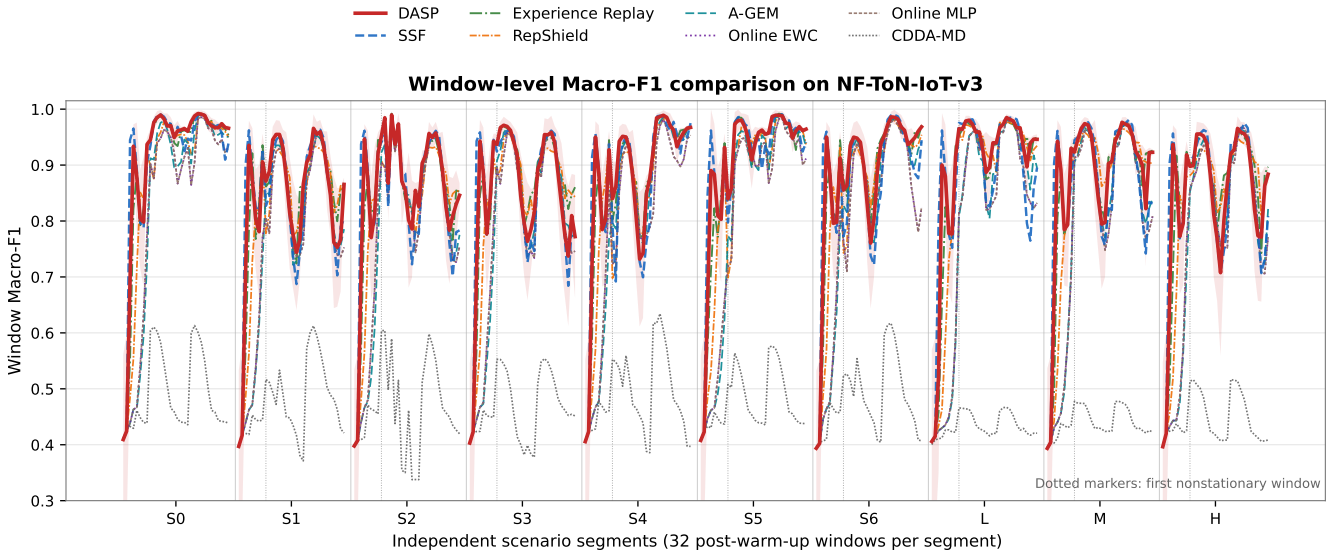


FIGURE 8. **Window-level Macro-F1 trajectories of all eight methods on NF-ToN-IoT-v3, averaged over five stream seeds and two model seeds. The frequent crossings between DASP and the strongest baselines visualize the smaller, condition-dependent effect on this dataset.**

reduces mean update time by approximately $1.90\times$ and gradient evaluations by approximately $1.56\times$. These observations establish implemented locality and lower adaptation work, but they do not isolate the causal contribution of each router, replay, or performance-gate component.

RQ3: Boundary conditions and systems trade-offs

The observed advantage weakens under gradual drift and high-intensity local drift, where attribution evidence is less sharply separated across modules. The strongest counterexample is the NF-BoT-IoT-v3/S2-Gradual cell with stream seed 2102 and model seed 91, where DASP trails SSF by 0.0900 Macro-F1. DASP also has substantially lower prediction throughput than the flat MLP-based alternatives because all four representation branches are evaluated before fusion. Localized repair is therefore most defensible when adaptation cost, bounded parameter motion, and auditability are prioritized; a prediction-throughput-constrained deployment may prefer a simpler or optimized architecture.

Mechanistic interpretation

The development sequence indicates that the earlier selective path was too conservative: classifier calibration alone did not provide enough historical context to stabilize benign and attack boundaries. Bounded rehearsal supplies that context, while the top-k mask prevents replay from becoming global

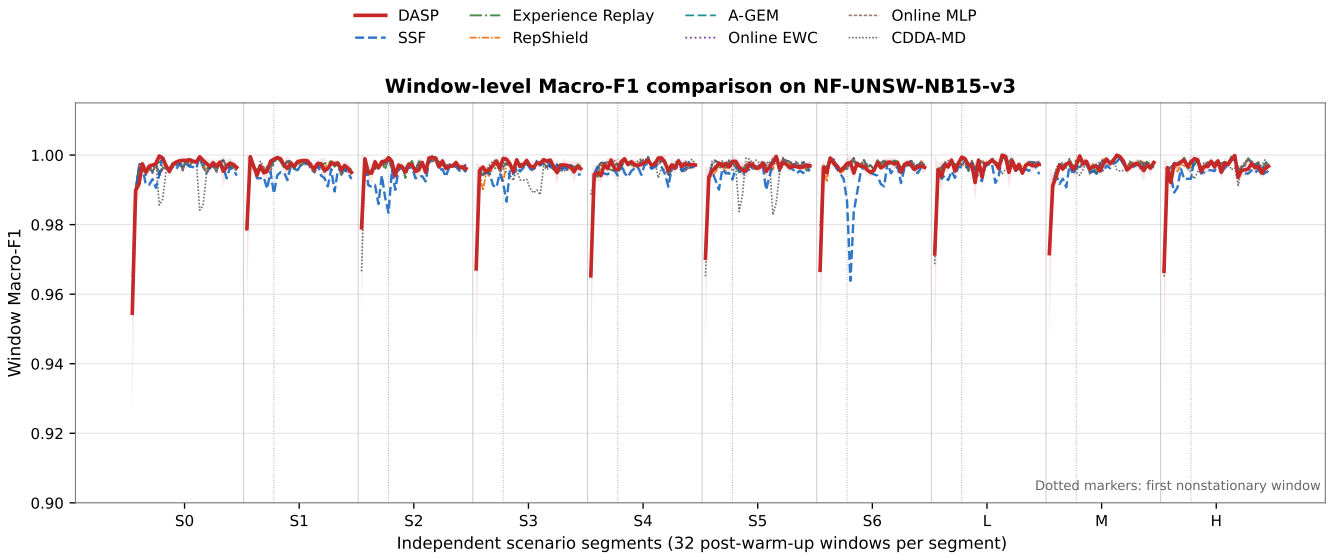


FIGURE 9. **Window-level Macro-F1 trajectories of all eight methods on NF-UNSW-NB15-v3. The compressed vertical scale reflects near-ceiling performance; differences among the leading methods are correspondingly small. DASP is emphasized in red and its band shows the 95% normal-approximation confidence interval.**

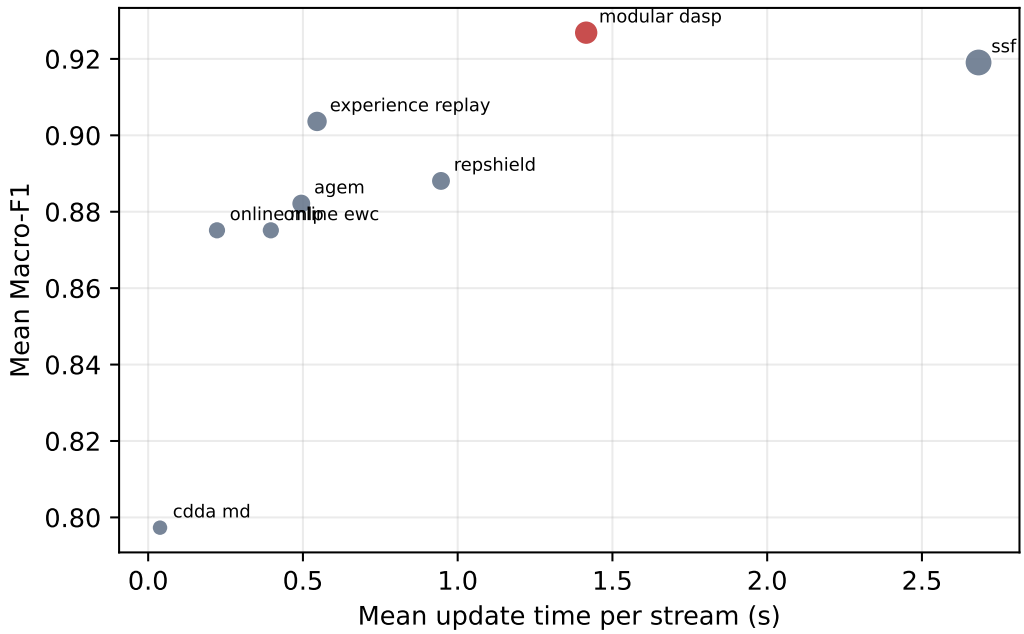


FIGURE 10. **Accuracy–adaptation frontier. Bubble size is proportional to gradient evaluations; DASP is highlighted in red.**

retraining. Because the ablation was sequential rather than factorial, this is a mechanism-consistent interpretation of the observed pattern rather than an isolated causal estimate for each component. The confirmation therefore validates the combined DASP system, not the independent predictive contribution of its attribution router.

Relation to interpretability

The router output is an intervention trace: "module 2 and module 4 were opened, modules 1 and 3 were frozen, and 512 historical examples were replayed." This explanation is operationally useful for incident response and model governance. It complements, rather than replaces, feature-level explanation methods such as LIME and SHAP [27]–[29].

VIII. THREATS TO VALIDITY AND LIMITATIONS

- The datasets are prepared chronological artifacts, not a live deployment. Their scenario transformations create controlled distributional changes but do not establish that every episode changes the real-world binary decision

function $P(Y \mid X)$. The term concept drift is therefore used for the experimental stress protocol, not as a claim that every transformed stream reproduces natural operational drift.

- The five stream seeds and two model seeds provide substantial replication but do not cover every possible preprocessing or architecture choice.
- The comparison is binary. Multi-class attack-family retention, open-set attacks, and severe label noise remain future work.
- The modular partition is feature-group based. Learned or attention-head routing may be preferable when features are not naturally ordered.
- The locked matrix does not include random top-k, fixed-module, MMD-only, or conflict-only routing controls. Consequently, the study does not identify how much of the observed performance is contributed by attribution-guided routing separately from replay, weighting, the modular backbone, and performance-gate logic.
- Primary DASP uses delayed repair-pool labels to select 128 class-balanced training examples and separately evaluates a 25% holdout; other paths use different selection semantics. The recorded 128 field is therefore not total label exposure. The matrix supports a protocol-specific comparison, while a strict shared-label contract requires rerunning all methods under one access rule.
- DASP's lower prediction throughput is a genuine systems limitation. The measured speed is hardware- and implementation-specific, so it should not be generalized as a universal throughput advantage. A production implementation should consider fused kernels, vectorized modules, or a sparse execution path.
- The confirmation tests are paired across cells; the window-level logs are used for diagnosis and not treated as independent statistical replications. The reported p-values therefore support the paired confirmation contrast under this protocol, not an unrestricted population-level claim.
- The primary comparator and method realizations are fixed by the locked study protocol. Unless an official implementation is explicitly identified, the results apply to these evaluated realizations rather than every possible realization of the named method. Independent reimplementation, external chronological data, and hardware replication remain necessary before broader claims about generality or production readiness.

IX. CONCLUSION

This paper presented DASP, a drift-attributed subspace-plasticity framework for streaming NIDS. DASP combines module-wise drift requirements, conflict-aware top-k routing, strict freezing, class-balanced bounded rehearsal, performance-gated local repair modes, and transparent localization logs. On the 4,000-run confirmation matrix, DASP achieved Macro-F1 0.9269 versus 0.9191 for SSF, won 383 of 500 paired cells, reduced update time by approximately 1.90-fold, and reduced gradient evaluations by approximately 1.56-fold. It performed no logged full-model or all-representation-module update, kept the shared classifier trainable, and never activated more than three of four representation modules in the repair records.

The integrated localized-repair system is a credible alternative under the implemented supervision paths, but the confirmation neither isolates the attribution router nor enforces equal total label exposure. The principal qualification is equally important: modular execution lowers prediction throughput and a small number of difficult drift cells remain. Future work should evaluate learned feature grouping, adaptive module budgets, natural chronological streams, multi-class attack-family retention, and controlled routing ablations comparing random top-k, fixed-module, MMD-only, conflict-only, and combined attribution policies under the same repair budget.

DATA AVAILABILITY

The archival repository contains the chronologically prepared streams, locked experimental configuration, executable method implementations, run-level confirmation records, and derived analysis outputs needed to reproduce the reported results. Public dataset provenance follows the sources cited in Section V.

ETHICS STATEMENT

This study uses public, de-identified network-flow benchmark data and performs no intervention on a live network. It involves no human participants, private records, or personal-contact data. Dataset use and redistribution must follow the source licenses.

AUTHOR CONTRIBUTIONS

The authors must replace this placeholder with a verified CRediT statement covering conceptualization, methodology, software, validation, formal analysis, data curation, visualization, writing, supervision, and project administration.

FUNDING

Funding information must be completed and verified by the authors. No funding source is asserted by this draft.

CONFLICT OF INTEREST

Conflict-of-interest information must be verified by every author. No competing-interest declaration is inferred by this manuscript preparation workflow.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

Generative AI tools assisted manuscript organization, language editing, method-implementation inspection, figure generation, and reproducibility checks. The authors remain responsible for verifying every numerical result, citation, methodological statement, and final claim.

X. DETAILED METHOD AND STRUCTURAL INVARIANTS

Section IV is the canonical method definition. This section retains only operational lifecycle details and structural prop-

erties needed to audit that definition; it does not introduce a second algorithm.

A. CALIBRATION AND REFERENCE LIFECYCLE

After warm-up, the evaluation procedure creates a frozen copy of the fitted model as the reference snapshot. The reference representation bank is extracted from the final warm-up window. The four calibration representation sets are then produced by that same frozen model on the last four warm-up windows, and each calibration MMD compares the fixed final-window bank with one calibration-window set. The robust median/MAD statistics are therefore based on a fixed reference, not leave-one-window-out calibration or independent resampling.

Reference refresh is enabled in the locked confirmation. After every applied primary DASP repair, the updated predictor is copied to form the new reference snapshot, the reference bank is recomputed on the delayed repaired window, and the recent-window calibration statistics are recomputed. The primary procedure has no candidate rejection or rollback stage.

B. DELAYED SUPERVISION AND REPAIR MODES

At outer iteration t, the online procedure predicts W_t, computes $\mathcal{E}_t$, and stores both in the pending queue. It then repairs $\tau = t - 1$ only after Y_τ arrives, using the stored $\mathcal{E}_\tau$ and newly computed module conflicts $c_{j,\tau}$. The labeled delayed window is split into a 75% repair pool and 25% holdout; repair-pool labels select at most 128 class-balanced training examples. Thus, 128 bounds training, not total label exposure.

The triggered mode uses positive top-k values of $p_{j,\tau} = r_{j,\tau}/(1 + c_{j,\tau})$ for two epochs. If MMD does not trigger, holdout Macro-F1 below 0.80 selects the three-epoch performance-gated emergency mode with plasticity $\min\{1, \max(0.25, c_{j,\tau})\}$; otherwise background local consolidation uses plasticity 0.25 for two epochs. Both no-trigger modes rank up to three modules by conflict and retain the deterministic fallback specified by the protocol. The classifier remains trainable at multiplier 1 on the base learning rate.

C. PROOF SKETCHES FOR STRUCTURAL PROPERTIES

The first property is inactive-module invariance. Let Θ_j be the parameter set of an inactive module and let the optimizer parameter list exclude Θ_j. No gradient update is applied to Θ_j. Since the weight-decay term is also implemented by the optimizer, it is not applied either. Therefore, conditional on no external state mutation, $\Theta_j^{t+1} = \Theta_j^t$. This is stronger than a soft regularizer, which only penalizes movement.

The second property is bounded plasticity. Since $\mathcal{A}_t = \mathrm{TopK}(a_t, k)$ and $k = 3 < M = 4$, at most three representation modules can be trainable in a repair. The classifier is shared and explicitly exempted from the module count. The implementation records the active set, so the bound can be verified from logs rather than inferred from the configuration.

The third property is bounded memory exposure. Each repair samples at most 512 replay records and at most 128 repair-training labels. The optimization set is therefore bounded by 640 examples before internal batching; the separate 25% delayed-label performance holdout is evaluated outside this training set. This prevents a long stream from silently increasing the per-window training set. The memory itself is capped at 1,024 records and maintained with class balancing.

D. COMPLEXITY ACCOUNTING

Let P_j be the parameter count of module j, P_c the classifier parameter count, B_r the replay size, B_n the new-label budget, and E_t the number of repair epochs. The trainable parameter count for a repair is

$$P_t = P_c + \sum_{j \in \mathcal{A}_t} P_j, \tag{15}$$

while the full model count is $P = P_c + \sum_j P_j$. The structural trainable fraction is $\rho_t = P_t/P$. The observed gradient-evaluation count additionally depends on the number of applied repairs, batch size, and emergency repairs. This is why the manuscript reports both a parameter-locality invariant and measured gradient evaluations. A low nominal parameter fraction would be insufficient if the router triggered many more epochs.

E. INTERPRETABILITY AS AN INTERVENTION RECORD

DASP does not claim that a selected module is a human-semantic feature such as ''port behavior'' without a separate feature audit. Its interpretability claim is narrower and testable: the method can report which parameterized subspaces were allowed to change for a particular drift episode. The record can be joined with the corresponding stream window, scenario label, repair mode, and post-repair metrics. This is closer to an intervention trace than to post-hoc attribution. Feature-level tools such as LIME and SHAP remain complementary because they explain individual predictions rather than the optimizer's repair scope.

XI. DATASET CONSTRUCTION AND STREAM SEMANTICS

A. DATASET ROLES

The five datasets serve different stress roles. CIC-IDS2017 provides enterprise traffic with several attack families and a commonly used reference point. NF-BoT-IoT-v3 stresses botnet and denial-of-service behavior in a NetFlow representation. NF-CSE-CIC-IDS2018-v3 provides a higher-performing but still heterogeneous enterprise/IoT setting. NF-ToN-IoT-v3 emphasizes telemetry-rich traffic and remains one of the harder datasets in the confirmation. NF-UNSW-NB15-v3 is a heterogeneous benchmark in which several methods approach a performance ceiling. These roles are not interchangeable: a method that wins on a saturated dataset may not be robust under a difficult gradual drift.

The prepared streams preserve the chronological order used by the experiment. Accompanying metadata identify the dataset, scenario, stream seed, feature count, window count, warm-up interval, and class statistics. The evaluation procedure processes one stream at a time and records a summary only after the prediction-first loop completes. No

summary from a future stream or model seed is visible to the online learner.

B. WINDOW PROTOCOL

Each confirmation stream is divided into 36 windows of 4,096 flows. Four warm-up windows establish the initial representation reference and classifier state. For a non-warm-up window W_t, the system first computes predictions and records the outputs. Only after the one-window delay are labels used for adaptation. Primary DASP splits the labeled delayed window 75/25 into repair and holdout partitions, then selects at most 128 class-balanced repair examples. The next window can therefore be influenced by W_t labels, but the prediction on W_t is never influenced by its own labels.

This ordering has three consequences. It increases realism relative to train-before-predict evaluation. It makes the delayed-supervision path an explicit part of the method contract. It also means that an early routing mistake can affect later windows, which is why the paired stream-level unit is more meaningful than treating each window as an independent test sample.

C. TEN SCENARIO DEFINITIONS

The primary evaluation includes one stationary control and nine drift conditions. S0 is the stable control stream and tests whether adaptation damages a model without a deliberate distribution change. S1-Sudden introduces an abrupt regime change. S2-Gradual spreads the transition across multiple windows. S3-Incremental applies a sequence of smaller changes. S4-Recurrent revisits a previously observed regime. S5-Blip inserts a transient episode followed by recovery. S6-Combined composes more than one drift pattern. S7-Local-Low, S7-Local-Medium, and S7-Local-High vary the intensity of a localized class-conditional change. The three local variants are especially important for DASP because they test whether increasing local severity expands the routed set or merely increases the update magnitude.

The complete dataset–scenario pairing is visualized in Fig. 11. This matrix is both a design diagram and an analysis map: every cell represents five stream seeds and two model seeds, and the color encodes the paired DASP–SSF Macro-F1 gap after aggregation. The figure therefore connects the controlled scenario taxonomy to the confirmation evidence without collapsing the study into a single mean.

D. SEED STRUCTURE AND PAIRING

The stream seeds are 2001, 2002, 2003, 2101, and 2102. The model seeds are 91 and 92. For a fixed dataset, scenario, stream seed, and model seed, all eight methods receive the same prepared stream, temporal delay, and initial seed assignment, but the audited label-selection and holdout semantics are method specific. The DASP–SSF comparison is therefore paired at the cell level. The 500 paired cells arise from five datasets, ten scenarios, five stream seeds, and two model seeds. The full 4,000-run count includes all eight methods; it is not a count of independent datasets.

TABLE 5. Primary scenario semantics and expected adaptation pressure.

Scenario	Operational interpretation
S0	Stationary control; penalizes unnecessary updates.
S1-Sudden	Abrupt regime replacement; tests fast detection and repair.
S2-Gradual	Slow transition; tests attribution ambiguity and delayed response.
S3-Incremental	Multiple small changes; tests repeated bounded repairs.
S4-Recurrent	Previously seen regime returns; tests retention and recovery.
S5-Blip	Short transient event; tests whether repair overreacts.
S6-Combined	Compound drift; tests interaction of mechanisms.
S7-Local-Low	Low-intensity local class-conditional change.
S7-Local-Medium	Medium-intensity local class-conditional change.
S7-Local-High	High-intensity local class-conditional change.

E. FAIRNESS CONTROLS

The methods are locked before the confirmation run. The warm-up length is four windows, the prepared-stream window size is 4,096, and CPU execution uses one thread. All paths record 128 new training examples per update, but label access differs: DASP uses labeled balanced selection plus a 25% holdout, SSF selects after label access, and budget-matched comparators query uniformly without labels. Total label exposure is therefore unequal. SSF retains its configured sample-selection and forgetting-protection strategy. Experience Replay, A-GEM, Online EWC, RepShield, Online MLP, and CDDA-MD are included because they represent replay, gradient projection, regularization, representation protection, direct online learning, and distributional detection families. DASP is not compared to a deliberately weakened SSF implementation; both use the same prediction-first delay, prepared streams, and seeds, while the supervision-accounting exception is disclosed separately.

XII. EVALUATION CONTRACT AND STATISTICAL ANALYSIS

A. PREDICTIVE METRICS

For a binary task, the per-window Macro-F1 is the arithmetic mean of the F1 score for the benign and attack classes. If $F_1^{(0)}$ and $F_1^{(1)}$ are the class-specific scores, then

$$\mathrm{MacroF1}_t = \frac{F_1^{(0)} + F_1^{(1)}}{2}. \tag{16}$$

Macro-F1 is the primary metric because accuracy can be dominated by benign traffic. Cumulative attack recall and cumulative false-negative rate are reported as operational safety measures. Online forgetting is computed by comparing historical behavior after each update with the corresponding pre-update reference. Window stability Macro-F1 summarizes the consistency of predictions across the stream rather than only the final aggregate.

B. COST AND PERFORMANCE METRICS

Adaptation time is measured as the sum of update intervals, excluding data loading and stream preparation. Median update time is reported because a mean can be dominated by emergency repairs. Gradient evaluations count the optimizer-equivalent forward/backward evaluations recorded during the

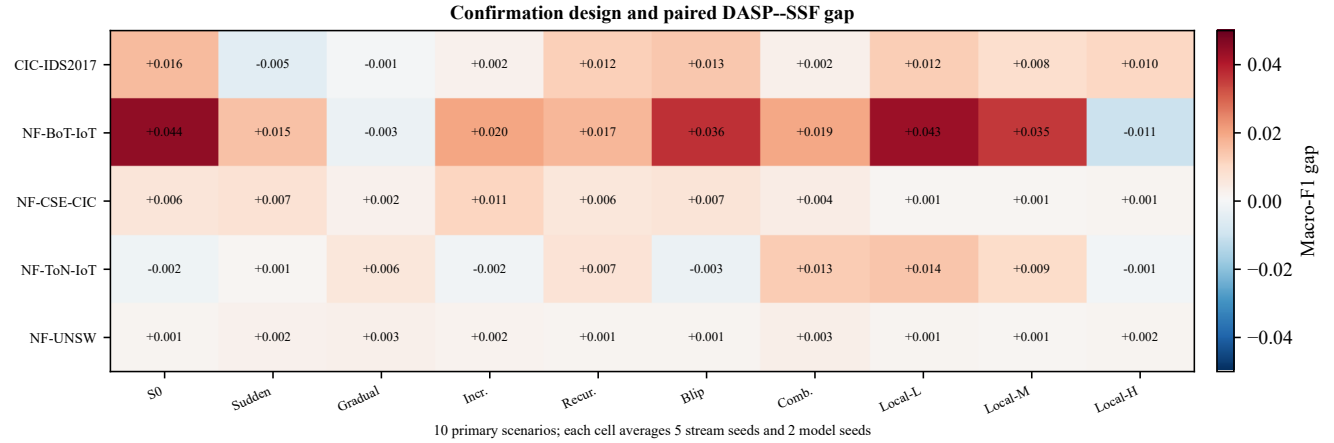


FIGURE 11. Confirmation design and paired performance map. Rows are datasets, columns are the ten primary scenarios, and each cell aggregates five stream seeds and two model seeds.

experiment. Prediction throughput is measured separately in flows per second. This distinction prevents a method from appearing efficient merely because it performs almost no adaptation. Memory payload, model parameter bytes, and peak resident-set increment are included to expose deployment costs.

C. PAIRED INFERENCE

Let $d_i = \mathrm{MacroF1}_i^{\mathrm{DASP}} - \mathrm{MacroF1}_i^{\mathrm{SSF}}$ for paired cell i. The primary summaries are $\bar{d}$, the median of d_i, the number of positive differences, and the number of cells within ± 0.01. The sign test evaluates whether the probability of a positive difference is one half. The paired t-test is used as a sensitivity analysis for the mean, while the Wilcoxon signed-rank test probes whether the signed-rank distribution is displaced from zero. These tests are not applied to individual windows because windows within one stream are serially dependent.

The confirmation result is therefore interpreted at the stream/model-seed cell level. A positive aggregate does not imply universal dominance. The worst cell is reported explicitly, and the scenario and dataset tables preserve the distributional context. This reporting choice is aligned with the goal of evaluating a deployment mechanism rather than selecting only the most favorable windows.

D. EFFECT SIZE AND UNCERTAINTY REPORTING

The mean gap is accompanied by the win count and sensitivity p-values. In a future archival release, the stored paired cell-level records can additionally support percentile bootstrap intervals over the 500 cells, with resampling performed at the paired-cell level. Resampling at the window level would understate uncertainty because it treats repeated observations from a single stream as independent. The present analysis keeps the exact confirmation values as the primary claim and treats any resampling interval as supplementary.

XIII. EXPANDED CONFIRMATION RESULTS

The compact result table in Section VI gives the headline comparison. The tables in this section expose the structure behind that headline. They are generated from the stored 4,000-row summary rather than manually transcribed. Consequently, readers can reproduce every displayed aggregate by grouping the public summary by dataset, scenario, and method.

A. DATASET-BY-METHOD PATTERN

Table 6 shows that DASP is not supported by a single favorable dataset. Its strongest practical separation from SSF occurs on NF-BoT-IoT-v3, where the mean gap is approximately $+0.0215$. The gap is smaller on NF-UNSW-NB15-v3 because several methods are near a ceiling. This pattern is important for interpretation: a high mean over all datasets could otherwise be driven by one easy benchmark. DASP also remains ahead on the harder NF-ToN-IoT-v3 setting, although the margin is modest.

The matrix shows a second pattern. Experience Replay is the strongest non-DASP baseline in aggregate, which is expected because replay supplies historical context without requiring a modular backbone. However, its update cost is lower partly because it uses a simpler update rule. DASP's contribution is not simply to minimize update time; it combines replay with an explicit constraint on where replay gradients can change the model. SSF remains the closest accuracy comparator and the appropriate primary baseline for the paired claim.

B. DATASET-LEVEL COST AND ACCURACY

Table 7 separates the accuracy gap from the systems ratio. DASP's update speedup remains above 1.7 across all five datasets, and the gradient reduction remains close to 1.5 or higher. The stable cost ratio suggests that the benefit is architectural rather than a consequence of one unusually short stream. The largest speedup appears on NF-UNSW-NB15-v3, the same dataset in which accuracy differences are compressed by the ceiling effect.

TABLE 6. Mean Macro-F1 by dataset and method over the confirmation matrix.

Dataset	DASP	SSF	ER	RepShield	A-GEM	Online MLP	Online EWC	CDDA-MD
cic-ids2017	0.9420	0.9350	0.9336	0.9284	0.9157	0.9109	0.9105	0.8672
nf-bot-iot-v3	0.8652	0.8437	0.7632	0.7181	0.7497	0.7302	0.7304	0.7349
nf-cse-cic-ids2018-v3	0.9627	0.9581	0.9600	0.9547	0.9521	0.9500	0.9500	0.9220
nf-ton-iot-v3	0.8684	0.8642	0.8651	0.8430	0.7970	0.7886	0.7886	0.4666
nf-unsw-nb15-v3	0.9961	0.9944	0.9962	0.9962	0.9962	0.9961	0.9962	0.9959

TABLE 7. Dataset-level DASP–SSF comparison. Speedup and gradient reduction are ratios relative to SSF.

Dataset	DASP F1	SSF F1	Gap	Wins/100	Update speedup	Grad. reduction
cic-ids2017	0.9420	0.9350	0.0070	78	1.86	1.59
nf-bot-iot-v3	0.8652	0.8437	0.0215	81	1.77	1.45
nf-cse-cic-ids2018-v3	0.9627	0.9581	0.0045	82	1.93	1.59
nf-ton-iot-v3	0.8684	0.8642	0.0042	47	1.90	1.58
nf-unsw-nb15-v3	0.9961	0.9944	0.0018	95	2.03	1.60

TABLE 8. Scenario-level DASP–SSF comparison across five datasets and ten replication cells per scenario.

Scenario	DASP F1	SSF F1	Gap	Wins/50	Update speedup	Grad. reduction
S0	0.9604	0.9473	0.0131	43	1.94	1.58
S1-Sudden	0.9196	0.9157	0.0040	34	1.89	1.57
S2-Gradual	0.9216	0.9202	0.0014	33	1.88	1.54
S3-Incremental	0.9273	0.9208	0.0065	38	1.89	1.56
S4-Recurrent	0.9256	0.9171	0.0085	41	1.91	1.57
S5-Blip	0.9430	0.9321	0.0110	39	1.91	1.58
S6-Combined	0.8902	0.8819	0.0083	38	1.86	1.53
S7-Local-Low	0.9359	0.9218	0.0141	42	1.92	1.55
S7-Local-Medium	0.9298	0.9188	0.0110	39	1.86	1.55
S7-Local-High	0.9152	0.9150	0.0002	36	1.90	1.54

TABLE 9. DASP Macro-F1 for every dataset–scenario cell. Values average five stream seeds and two model seeds.

Dataset	S0	S1	S2	S3	S4	S5	S6	L-Low	L-Med.	L-High
cic-ids2017	0.9623	0.9332	0.9344	0.9388	0.9292	0.9476	0.9225	0.9484	0.9539	0.9496
nf-bot-iot-v3	0.9388	0.9115	0.8998	0.9068	0.8955	0.9099	0.7092	0.8504	0.8330	0.7970
nf-cse-cic-ids2018-v3	0.9919	0.9218	0.9294	0.9508	0.9373	0.9636	0.9476	0.9958	0.9951	0.9932
nf-ton-iot-v3	0.9131	0.8352	0.8479	0.8440	0.8699	0.8980	0.8759	0.8888	0.8706	0.8402
nf-unsw-nb15-v3	0.9958	0.9965	0.9965	0.9961	0.9961	0.9959	0.9958	0.9962	0.9962	0.9959

The cost table also prevents an overly broad claim. DASP does not improve every systems metric. Its forward throughput is lower because four module branches and a fusion stage are executed for every prediction. The appropriate conclusion is therefore a two-axis one: DASP improves the accuracy–adaptation trade-off while paying a forward-inference cost. A deployment that is constrained by prediction latency rather than adaptation latency may prefer a fused or sparse implementation in future work.

C. SCENARIO-LEVEL RESPONSE

Table 8 reports all ten primary conditions. The largest mean gains occur in S0, S7-Local-Low, S5-Blip, and S7-Local-Medium. The stationary control result is particularly informative: a positive S0 gap indicates that the performance-gated modular repair path does not merely damage the baseline model when the stream is stable. The local-low and blip gains are consistent with the intended use case for attribution, namely, a change confined to a subset of the representation or a short-lived episode. They should be read as scenario-level evidence, not as proof that every local change is correctly attributed.

S2-Gradual and S7-Local-High are weaker cases. Gradual drift gives the router a less distinct boundary between reference and current representations. High-intensity local drift can produce evidence in several modules, reducing the advantage of a strict top-k selection. These conditions motivate adaptive routing budgets and uncertainty-aware expansion, but they do not invalidate the primary result because the confirmation configuration was locked before inspecting the final summary.

D. CELL-LEVEL MATRICES

Tables 9–11 provide a compact audit trail. The DASP and SSF matrices show whether a dataset's aggregate is driven

TABLE 10. SSF Macro-F1 for every dataset–scenario cell, using the same paired cells as DASP.

Dataset	S0	S1	S2	S3	S4	S5	S6	L-Low	L-Med.	L-High
cic-ids2017	0.9464	0.9383	0.9354	0.9366	0.9172	0.9343	0.9200	0.9363	0.9456	0.9393
nf-bot-iot-v3	0.8946	0.8969	0.9026	0.8870	0.8788	0.8736	0.6899	0.8077	0.7977	0.8080
nf-cse-cic-ids2018-v3	0.9857	0.9147	0.9272	0.9401	0.9317	0.9570	0.9439	0.9950	0.9941	0.9918
nf-ton-iot-v3	0.9152	0.8342	0.8421	0.8458	0.8630	0.9008	0.8631	0.8750	0.8613	0.8416
nf-unsw-nb15-v3	0.9944	0.9943	0.9938	0.9944	0.9948	0.9945	0.9927	0.9950	0.9952	0.9944

TABLE 11. DASP minus SSF Macro-F1 gap for every dataset–scenario cell. Positive values favor DASP.

Dataset	S0	S1	S2	S3	S4	S5	S6	L-Low	L-Med.	L-High
cic-ids2017	0.016	-0.005	-0.001	0.002	0.012	0.013	0.002	0.012	0.008	0.010
nf-bot-iot-v3	0.044	0.015	-0.003	0.020	0.017	0.036	0.019	0.043	0.035	-0.011
nf-cse-cic-ids2018-v3	0.006	0.007	0.002	0.011	0.006	0.007	0.004	0.001	0.001	0.001
nf-ton-iot-v3	-0.002	0.001	0.006	-0.002	0.007	-0.003	0.013	0.014	0.009	-0.001
nf-unsw-nb15-v3	0.001	0.002	0.003	0.002	0.001	0.001	0.003	0.001	0.001	0.002

TABLE 12. Operational and retention metrics by method. Values are means over 500 confirmation cells.

Method	Attack recall	FNR	Forgetting	Stability F1	Update labels	Peak RSS inc. (MB)	Model bytes (KB)
DASP	0.9591	0.0409	0.0473	0.9587	4,096	28.6	82.0
SSF	0.9569	0.0431	0.0572	0.9511	4,096	28.9	82.0
Experience Replay	0.9384	0.0616	0.0544	0.9634	4,096	28.4	82.0
RepShield	0.9304	0.0696	0.0589	0.9650	4,096	30.1	82.0
A-GEM	0.9271	0.0729	0.0569	0.9635	4,096	29.1	82.0
Online MLP	0.9215	0.0785	0.0607	0.9616	4,096	31.1	82.0
Online EWC	0.9214	0.0786	0.0607	0.9616	4,096	28.9	82.0
CDDA-MD	0.8961	0.1039	0.0800	0.9710	4,096	27.9	82.0

TABLE 13. Logged DASP structural invariants.

Invariant	Value
Confirmation cells	500
Repair records	16,000
Replay examples per repair	512
Global update count	0
Maximum active modules	3 of 4
Median active modules	3 of 4
Minimum active modules	1 of 4
Memory capacity	1,024

by a single scenario. The gap matrix shows that positive and negative cells coexist even when the scenario mean is positive. The negative NF-BoT-IoT-v3/S2-Gradual cell is therefore not removed as an outlier. It is part of the mechanism's empirical boundary and should be used to design the next ablation.

E. RETENTION AND OPERATIONAL BEHAVIOR

Table 12 adds metrics that are not visible in Macro-F1 alone. Attack recall and false-negative rate matter because a detector can obtain a reasonable mean F1 while missing a minority attack class. Forgetting and stability show whether a repair creates a short-term gain at the cost of oscillation. Memory and resident-set measurements distinguish a bounded replay algorithm from a method whose history grows with the stream.

DASP's structural invariant is summarized in Table 13. Across the 500 confirmation cells, 16,000 repair records were available. No logged repair performed a global update. The maximum active representation-module count was three of four, the median was three, and the minimum was one. These values should be read together with the classifier exception: the shared classifier is always an explicit adaptation group, while the invariant concerns the four representation modules.

XIV. ABLATION, SENSITIVITY, AND FAILURE DIAGNOSTICS

A. DEVELOPMENT ABLATION AS MECHANISM DIAGNOSIS

The development pilot was not used to tune the confirmation statistics, but it is useful for understanding why the final mechanism contains several components. The initial no-update path reached mean Macro-F1 0.7625 against SSF at 0.9211. This is a failure of under-adaptation. Classifier calibration raised the score to 0.8706, and the early semi-local prototype reached 0.8725. The large jump to approximately 0.9266 occurred only after bounded localized rehearsal and new-sample weighting were introduced. The sequence suggests that locality alone is not sufficient: a selective model still needs enough historical and current supervised context to stabilize the boundary.

The ablation should not be interpreted as a factorial causal estimate because the pilot variants were developed sequentially. It does, however, reject a tempting but incorrect simplification: simply freezing modules and adding a drift score is not enough. The successful mechanism combines attribution, bounded replay, weighted delayed labels, strict optimizer masking, and the delayed-label performance gate.

TABLE 14. Development ablation sequence on the 15 paired pilot cells.

Variant	Macro-F1	Interpretation
No update	0.7625	Under-adaptation
Classifier calibration	0.8706	Shared boundary repair
Early semi-local prototype	0.8725	Partial locality without sufficient rehearsal
Localized rehearsal and new-sample weighting	0.9266	Final mechanism direction
SSF comparator	0.9211	Development reference

B. ROUTER SENSITIVITY

The locked primary router uses three kernel scales, persistence one, a severity threshold of 0.15, and top-$k = 3$. The emergency-severity value of 0.60 in the locked experimental configuration is inactive because it is not used by the evaluated primary DASP procedure. Each setting encodes a trade-off. Increasing the kernel scale set can improve sensitivity to broad representation shifts but increases the per-window discrepancy calculation. Raising persistence can suppress false positives but delays repair under sudden drift. A smaller k increases structural sparsity but risks leaving the classifier without a path to the changed representation. A larger k improves recoverability but moves DASP toward global replay.

Because the primary confirmation was locked, the correct use of sensitivity analysis is prospective. The saved logs should be replayed under $k \in \{1, 2, 3, 4\}$, persistence values in $\{1, 2\}$, and alternative attribution weights $\alpha \in \{0.25, 0.5, 0.75\}$. The comparison should report accuracy, update time, active-module fraction, and worst-cell degradation. In particular, $k = 4$ should be treated as a locality ablation rather than as a production configuration, because it removes the principal structural constraint.

C. FAILURE EPISODE ANALYSIS

The worst confirmation cell occurs on NF-BoT-IoT-v3 under S2-Gradual with stream seed 2102 and model seed 91, where DASP trails SSF by 0.0900 Macro-F1. The failure is not hidden by reporting only dataset means. A plausible mechanism is attribution ambiguity: the gradual change produces moderate evidence in several modules before the classifier has a stable boundary, and the top-k repair allocates limited plasticity among modules whose evidence is not yet separable.

The failure suggests three concrete follow-up diagnostics. First, plot the four attribution trajectories against the delayed attack recall to determine whether the router leads or lags the performance drop. Second, compare the selected set with an oracle set computed from a post-hoc feature perturbation analysis; this tests attribution quality without changing the primary result. Third, evaluate a temporary uncertainty-triggered expansion that opens the fourth module only when the delayed-label performance gate falls below threshold twice consecutively. Such a diagnostic must be labeled as a new method, not retroactively applied to the confirmation run.

D. NEGATIVE CONTROLS

S0 is a negative control for unnecessary plasticity. If DASP had a strong negative S0 gap, the method would be overreacting to noise. The observed positive S0 mean gap, together with the performance-gated maintenance mechanism, provides evidence against that failure mode. A second negative control is the global-update counter: a method could claim locality while silently updating every module. The recorded global-update count of zero and the active-module upper bound address this implementation risk.

XV. THEORETICAL AND OPERATIONAL ANALYSIS

A. CONSTRAINED REPAIR VIEW

Let $\tau = t - 1$ be the delayed window whose stored evidence is restored at outer iteration t. The pre-update mode is selected by

$$m_\tau = \pi\left(s_\tau, F_\tau^{\mathrm{hold}}\right), \tag{17}$$

where s_τ is stored MMD severity and F_τ^{hold} is the delayed-label performance-holdout Macro-F1. Given the routed set $\mathcal{A}_\tau$ and mode-specific feasible update set $\Omega(\mathcal{A}_\tau, m_\tau)$, the implemented optimization is represented by

$$\Delta_\tau = \arg \min_{\Delta \in \Omega(\mathcal{A}_\tau, m_\tau)} \mathcal{L}_{\mathrm{new}}(\Theta + \Delta) + \lambda_r \mathcal{L}_{\mathrm{replay}}(\Theta + \Delta), \tag{18}$$

subject to

$$\Delta_j = 0 \quad \forall j \notin \mathcal{A}_\tau, \qquad |\mathcal{A}_\tau| \le k, \qquad |B_\tau^{\mathrm{mem}}| \le 512. \tag{19}$$

The holdout loss is not part of the training objective and does not accept or reject a post-update candidate. It selects emergency versus background intensity before optimization. Replay constrains forgetting through data, whereas optimizer exclusion imposes exact inactivity on unrouted representation modules.

The Lagrangian perspective also clarifies why a soft penalty is not equivalent to DASP. A penalty such as $\lambda \|\Theta_j - \Theta_j^{\mathrm{old}}\|^2$ may reduce movement but does not guarantee $\Delta_j = 0$. DASP uses an equality constraint implemented through optimizer parameter exclusion. This makes the locality claim falsifiable: a parameter hash comparison can either confirm or reject the invariant.

B. LOCALITY VERSUS EXPRESSIVITY

The top-k constraint creates an explicit accuracy–locality trade-off. If $k = 1$, the update is highly selective but may underfit a compound drift. If $k = 4$, all representation modules can change and the structural distinction from global replay becomes weak. The locked value $k = 3$ is a middle point that permits recovery when several modules carry evidence while preserving at least one frozen representation path. The

confirmation results should therefore not be generalized to arbitrary k without a new sensitivity experiment.

The classifier exception creates a second trade-off. A classifier-only repair is computationally cheap and can correct a shifted class prior, but it cannot create a new representation for an unseen attack pattern. A module-only repair can change the representation but may leave the decision boundary miscalibrated. DASP updates the classifier explicitly while routing representation plasticity, which is why the method is best described as localized subspace repair rather than isolated module retraining.

C. ATTRIBUTION STABILITY UNDER DRIFT TYPES

The router is expected to behave differently across the ten scenarios. Under S1-Sudden, the discrepancy trajectory should change sharply and the delayed conflict should become concentrated after one label delay. Under S2-Gradual, the signals may rise slowly and become less separable across modules. Under S4-Recurrent, historical memory may reduce conflict if the returning regime is already represented. Under S5-Blip, the delayed-label performance holdout may escalate the no-trigger path when transient performance falls below its threshold. Under the local variants, the key question is not only whether a score is large, but whether the ranking of modules remains stable enough for top-k selection.

A useful diagnostic is the rank persistence of the routed set. Let $\mathcal{A}_t$ and $\mathcal{A}_{t+1}$ be two consecutive active sets. Their overlap can be measured by

$$J_t = \frac{|\mathcal{A}_t \cap \mathcal{A}_{t+1}|}{|\mathcal{A}_t \cup \mathcal{A}_{t+1}| + \epsilon}. \tag{20}$$

High overlap can indicate a persistent localized drift, while low overlap can indicate either a compound drift or an unstable router. The current confirmation summary records active-module counts but does not claim a causal result about rank persistence. This statistic is a direct extension for the next diagnostic release.

D. WHY THE PERFORMANCE HOLDOUT IS NOT AN ORACLE

The delayed-label performance holdout uses labels unavailable at prediction time, but it is not an oracle for the future stream. In the locked modular path it measures bounded delayed-window performance and chooses emergency or background intensity when MMD does not trigger. A favorable score can still be followed by an unseen regime, so the gate is not a guarantee of future optimality.

The prediction-first asymmetry remains: mode selection cannot reverse a prediction that has already been emitted on the current window. A deployed system should therefore pair the logged local update with monitoring of subsequent windows and an external model registry if operational rollback is required.

E. COMPUTATIONAL DECOMPOSITION

The per-window computation has four components: representation extraction, discrepancy estimation, delayed-label subset construction, and local repair. Representation extraction is shared with prediction and scales with the full input batch. Module-wise MMD uses a bounded sample of 128 representations and three kernel scales. The repair term scales with 512 replay examples and at most 128 repair-training labels; holdout evaluation is separate. Performance-holdout evaluation is used on the no-trigger path to select repair intensity. The top-k mask reduces the trainable parameter set but does not remove the cost of computing routing evidence. This explains why DASP improves adaptation time while reducing forward throughput relative to simpler flat MLP baselines.

The evidence map is deliberately conservative. It distinguishes a property guaranteed by the optimizer structure from a property merely suggested by aggregate performance. For example, zero global updates is a structural property of the evaluated procedure, whereas the claim that attribution causes the accuracy gain requires the prospective causal router ablation described in the limitations.

XVI. DEPLOYMENT THREAT MODEL AND GOVERNANCE

A. OPERATIONAL DEPLOYMENT BLUEPRINT

B. THREATS AGAINST ADAPTATION

An adaptive NIDS has threats that a static classifier does not. An attacker may attempt to induce false drift by generating a short burst of unusual but benign-looking traffic. A compromised labeling process may inject incorrect delayed labels and push the router toward an incorrect module. A replay-memory poisoning attack may insert mislabeled or atypical examples. A resource-exhaustion attack may increase the frequency of drift-like windows and cause repeated scoring and repair. DASP does not claim to solve these threats automatically, but its bounded and logged design makes them explicit monitoring targets.

C. GOVERNANCE AND CHANGE CONTROL

A production commit should carry a model version, stream identifier, active-module set, attribution vector, performance-holdout scores, replay cardinality, optimizer configuration, and a hash of inactive parameters. The record should be immutable after commit. If a later incident reveals that a repair was harmful, the operator should be able to reconstruct the candidate, identify the delayed labels used, and restore the previous model version. This is stronger than storing only a final checkpoint because the checkpoint does not explain why a change was made.

The same governance layer supports controlled experimentation. A new router threshold should be deployed as a versioned configuration and evaluated in shadow mode before it is allowed to commit. The primary confirmation configuration should remain immutable, while prospective variants use distinct run and method identifiers. This separation prevents an apparently small operational change from contaminating the published baseline.

TABLE 15. Mechanism-to-evidence mapping used in the evaluation.

Claim	Implementation evidence	Empirical evidence
Drift is localized	Four explicit feature modules and per-module scores	Module-count distribution and routing logs
Inactive modules are protected	Optimizer exclusion and parameter-hash comparison	Zero global updates; maximum three active modules
Replay stabilizes repair	Class-balanced 512-example historical batch	Development ablation and retention metrics
Adaptation training is bounded	At most 128 new training examples plus disclosed labeled holdout; memory 1,024; top-$k = 3$	Gradient evaluations and update-time distribution
Repair mode is explicit	Performance-holdout Macro-F1 and fixed emergency/background rules	Mode logs and worst-cell analysis
DASP improves the primary comparator	Paired cell-level confirmation design	383/500 wins and positive statistical tests

TABLE 16. Threats to the adaptive repair path and corresponding controls.

Threat	DASP control or required deployment control
False drift injection	Persistence, severity threshold, and performance-holdout monitoring
Label poisoning	Label provenance, class-balance checks, and external checkpoint rollback
Replay poisoning	Bounded memory, provenance fields, and periodic memory audit
Repair flooding	Cooldown, update budget, and rate-limited repair service
Silent global update	Optimizer exclusion, parameter hashes, and global-update counter
Audit tampering	Append-only signed routing and model-state records

D. EVIDENCE INTEGRATION AND SCOPE

The evidential argument should connect mechanism, protocol, result, and limitation rather than rely on one headline score. In this study, the mechanism is the routed subspace and strict optimizer mask. The protocol is the prediction-first delayed-label evaluation across five datasets and ten scenarios. The result is the paired accuracy and adaptation-cost advantage. The principal boundaries are the throughput penalty and the difficult gradual-drift cell. Reporting these components together allows the contribution to be evaluated within its empirical scope.

XVII. EXPERIMENTAL INTEGRITY AND REPRODUCIBILITY CHECKS

A. LEAKAGE AUDIT

The most damaging error in a continual-learning experiment is not a slightly suboptimal hyperparameter; it is information leakage. DASP's protocol prevents current-window leakage by emitting predictions before delayed-label access. The same order is enforced for every baseline. The delayed label queue is indexed by window identifier, so a label batch can only be consumed after its configured delay. Stream metadata remain fixed during evaluation, and the independent analysis workflow operates only on completed run records rather than feeding aggregate statistics back into the online procedure.

The leakage audit also covers preprocessing. Standardization statistics are resolved from the training or warm-up portion of the prepared stream, not from the concatenation of all windows. If a future window were allowed to influence the scaler, the resulting performance would not represent a prediction-first deployment. The reproducibility record therefore includes feature count, warm-up length, and the locked configuration alongside the summary metrics.

B. BASELINE EQUIVALENCE AUDIT

The DASP–SSF comparison is paired, but pairing alone does not guarantee fairness. The equivalence audit verifies identical streams, seeds, window size, delay, and recorded training-subset count, while explicitly treating DASP–SSF holdout and total label exposure as unequal. A missing or partial run is not replaced with a zero and is not silently dropped from only one method.

A second fairness question concerns computational budgets. DASP uses a modular router and replay batch, while SSF uses its own strategic sample-selection mechanism. The correct comparison is not to force identical internal operations; it is to hold the externally meaningful constraints fixed and report internal work explicitly. The manuscript therefore reports gradient evaluations, update seconds, throughput, memory, and accuracy together. This allows a reader or system operator to decide whether the trade-off is favorable for a particular deployment.

C. RESULT INTEGRITY AUDIT

The result integrity audit recomputes the headline Macro-F1 and update-cost values from the row-level records. Displayed values are generated independently rather than copied from the manuscript text. The paired comparison is joined on dataset, scenario, stream seed, and model seed. The sign-test, paired t-test, and Wilcoxon values are calculated from the same 500 paired differences. This chain prevents a common failure mode in large experiments, where a figure and a table are produced from different filtering rules.

D. INTERPRETATION OF STATISTICAL SIGNIFICANCE

The extremely small paired p-values indicate that the direction of the aggregate comparison is unlikely to arise from a symmetric zero-median difference under the stated tests. They do not establish that DASP is superior on every dataset, every scenario, or every future deployment. The paired effect size is modest in Macro-F1 units, and its practical importance depends on the cost of false negatives and on the value assigned to adaptation time. The paper therefore reports both the mean gap and the operational trade-off rather than presenting significance as a substitute for engineering relevance.

TABLE 17. Experimental integrity checks and the artifact supporting each check.

Evaluation question	Check performed	Supporting artifact
Was the current label used before prediction?	Inspect prediction-first procedure and delay queue	Protocol specification and stream summaries
Were DASP and SSF paired fairly?	Join on dataset/scenario/stream/model seed	Paired cell-level comparison
Is locality structural or cosmetic?	Exclude inactive parameters from optimizer and verify fingerprints	DASP routing records
Can the replay memory grow without bound?	Check fixed capacity and fixed repair sample	Locked configuration and run summaries
Are negative cases hidden?	Report minimum gap and worst cell	Independent failure-case analysis
Is the throughput cost visible?	Report prediction throughput separately from update time	Method-level summary
Can the figures be regenerated?	Regenerate from row-level summaries	Independent figure-generation workflow

E. FAILURE TRANSPARENCY

A credible evaluation should expose the strongest counterexample. DASP's worst confirmation cell is retained with its dataset, scenario, stream seed, and model seed. The failure is not explained away as random noise, and no post-hoc change is applied to the primary result. Instead, the failure motivates a future adaptive-k and uncertainty-triggered expansion study. This preserves the separation between confirmatory evidence and prospective improvement.

F. TEST COVERAGE AND REPRODUCIBILITY GATE

The implementation is evaluated in layers rather than only through the final benchmark. Unit-level checks cover the memory sampler, class balancing, delayed-label queue, requirement normalization, top-k mask, optimizer exclusion, performance-gate thresholding, and summary recording. Integration checks cover one complete prediction-first stream, including background and emergency local modes. Benchmark checks cover the full pilot and confirmation matrices. This layered structure is important because a high-level Macro-F1 score cannot detect a silent global update or a label-ordering error.

The reproducibility gate has four stages. Stage one checks that the required numerical libraries are available and that deterministic one-thread execution is enabled. Stage two runs a small-scale evaluation and compares the resulting record structure with the locked protocol. Stage three recomputes the aggregate tables from row-level records and verifies that the paired join contains exactly 500 cells. Stage four compiles the manuscript and checks that all citations, references, figures, and generated tables resolve. A release is not considered reproducible if only the PDF can be compiled while the result summaries cannot be regenerated.

The existing test result of 95 passed and one skipped should be reported with its environment context. A future archival package should pin the numerical-computing framework versions, record the operating system and CPU model, and preserve the locked experimental configurations. Any numerical compatibility warning observed in the local environment should be removed or documented before public release because a warning that changes tensor behavior would threaten reproducibility even if the final table happened to look plausible.

A second gate concerns analysis independence. The independent analysis workflow may read completed summaries, but it must not rewrite method outputs or alter paired cells after inspecting the results. If an error is found, the correction must be applied to the evaluation procedure or the data-preparation stage, followed by a new full run. This rule prevents a common form of accidental p-hacking in which an inconvenient cell is removed from the summary rather than diagnosed at the source.

The final gate is visual. The compiled PDF should be rendered at ordinary resolution and inspected for figure readability, caption association, table continuation, reference clipping, and whitespace caused by floats. Figures that are mathematically correct but unreadable at column width should be regenerated with fewer labels or moved to a two-column layout. The figures are therefore separated into mechanism diagrams and result matrices rather than compressed into one dense composite figure.

G. EVIDENCE AND REPRODUCIBILITY BOUNDARY

Taken together, the integrity checks connect the method contract, prediction-first evaluation procedure, row-level records, paired analysis, and compiled manuscript. They reduce the risk of leakage, incomplete pairing, and silent global updates, but they do not substitute for external replication. An archival release should therefore include verified dataset licenses, resolved bibliographic metadata, a pinned software environment, and an author-independent rerun from the released configuration.

XVIII. OPERATIONAL REPRODUCIBILITY AND AUDITABILITY

The experiment is reproducible at three levels. At the protocol level, stream metadata specify the 4,096-flow window size and dataset-specific feature counts, while the locked experimental configuration specifies warm-up length, seeds, label delay, memory capacity, and router thresholds. At the run level, each experimental cell is uniquely indexed by dataset, scenario, method, stream seed, and model seed. At the analysis level, row-level and aggregate records contain the values used by the manuscript tables and figures.

A reproduction should verify the following invariants before comparing accuracy. The protocol metadata must specify 36 chronological 4,096-flow windows and 78 CIC-IDS2017 or 49 NF-v3 features. Prediction must precede delayed-label access, and supervision accounting must separate the 128-example training subset from total label exposure; a strict shared-label confirmation requires rerunning. The model seed must be reset

TABLE 18. Layered software and experiment validation gate.

Layer	Representative checks	Release criterion
Unit	Memory capacity, class balance, mask cardinality, queue delay	All deterministic unit tests pass
Model	Module output dimensions, classifier fusion, parameter hash	Inactive modules remain unchanged
Router	MMD calibration, conflict normalization, top-k selection	Scores and masks are recorded
Mode control	Performance-holdout threshold, background mode, emergency mode	Selected mode and locality fields are recorded
Evaluation procedure	Seed reset, prediction-before-label order, record structure	Small-scale evaluation completes with expected fields
Pilot	1,200-run matrix and method coverage	Every dataset/scenario/method cell is present
Confirmation	4,000-run matrix and paired join	Exactly 500 DASP–SSF paired cells
Analysis	Tables, figures, tests, quantiles, failure cell	Values recompute from row-level outputs
Manuscript	BibTeX, labels, figures, PDF compilation	Zero fatal/citation/reference errors

TABLE 19. Reproducibility checklist for a confirmation cell.

Check	Expected value
Dataset and scenario	Protocol identifier
Stream seed	2001, 2002, 2003, 2101, or 2102
Model seed	91 or 92
Window size	4,096 flows
Warm-up windows	4
Label delay	1 window
New-example training subset	At most 128 examples/window; not total label exposure
DASP memory capacity	1,024 records
DASP replay sample	512 records/repair
Maximum active modules	3 of 4
Global update count	0 for DASP
Prediction ordering	Before current labels

per cell. Each run must record update time separately from prediction throughput. Finally, the structural record must show that inactive module parameters are unchanged across a logged repair.

The audit trail is also useful for security operations. For each repair, an analyst can ask what changed, why the router selected those modules, which performance-gated mode was selected, and whether historical behavior was preserved. This does not replace a formal change-management process, but it makes the adaptation decision inspectable. In a production system, the same record should be signed or stored in an append-only log so that the model state and the explanation cannot be separated after an incident.

APPENDIX A LOCKED CONFIGURATION

The evaluated DASP configuration uses a four-module parallel MLP with module width 64, a 128-unit batch size, two epochs per triggered/background repair, AdamW learning rate 10^{-3}, weight decay 10^{-4}, one-window label delay, a class-balanced optimization subset of at most 128 examples selected from the labeled repair pool, a separate disjoint 25% labeled performance holdout, memory capacity 1,024, and 512 replay examples for DASP repair. The primary router uses multi-scale kernel scales $\{0.5, 1, 2\}$, severity threshold 0.15, persistence 1, and $k = 3$; the emergency-severity value 0.60 in the locked configuration is inactive in the evaluated primary procedure. New delayed-label examples receive weight 3.0; background modules use plasticity 0.25; emergency modules use $\min\{1, \max(0.25, c_j)\}$ and three epochs. The classifier uses multiplier 1 on the base learning rate, while all inactive modules are excluded from the optimizer. Reference refresh is enabled and runs after every applied primary DASP repair.

APPENDIX B EXPERIMENTAL PROTOCOL SUMMARY

The full pilot contains 1,200 runs: five datasets, ten scenarios, three stream seeds, one model seed, and eight methods. The confirmation contains 4,000 runs: five datasets, ten scenarios, five stream seeds, two model seeds, and eight methods. Before archival release, the software tests and environment-compatibility checks must be repeated using the final released materials; this manuscript therefore does not assert a current test-pass count. Any numerical-computing compatibility warning must be resolved or documented before independent reproduction.

APPENDIX C COMPLETE METHOD CONFIGURATION MATRIX

This appendix records the configuration contract used by the confirmation protocol. All methods receive the same prepared stream, four-window warm-up, one-window label delay, and the same model and stream seeds. The audit identified method-specific supervision semantics: all locked paths record 128 new training examples per update, but primary DASP constructs that subset from a labeled repair pool and additionally evaluates a disjoint 25% labeled performance holdout; primary SSF also selects from a delayed labeled pool, while several comparator paths use label-blind uniform selection. Therefore, the current matrix does not establish a strict shared total-label-exposure contract.

The matrix clarifies a potential source of confusion in comparative studies. DASP's trainable unit is not equivalent to using a smaller batch. DASP still evaluates the full modular predictor at prediction time and still uses historical examples during a repair. The distinction is where gradients are allowed to change parameters. Similarly, a baseline with fewer gradient evaluations is not automatically more efficient in the operational sense if it obtains substantially lower recall or performs almost no adaptation.

A. LOCKED DASP VALUES

The final DASP configuration uses four modules, module width 64, hidden dimensions 128 and 64 in the surrounding

TABLE 20. External evaluation controls and the audited DASP supervision exception.

Control	Value	Rationale
Task	Binary benign/attack	Common label space for all methods
Input features	78 for CIC-IDS2017; 49 for NF-v3 streams	Dataset-specific fixed representation interface
Window size	4,096 flows	Identical temporal granularity
Warm-up	4 windows	Initial reference and classifier calibration
Label delay	1 window	Prediction-first deployment constraint
Recorded new training examples	128/window	Common optimization-subset count, not common total label exposure
DASP repair-training subset	At most 128/window	Class-balanced selection from the labeled repair pool
Stream seeds	2001, 2002, 2003, 2101, 2102	Replication over stream construction
Model seeds	91, 92	Replication over initialization
Device	CPU, one thread	Deterministic performance measurement
Memory capacity	1,024 records	Bounded historical state
DASP performance holdout	0.25 of delayed window	Labeled pre-update emergency/background mode selection; outside the 128-example training subset
Primary metric	Macro-F1	Balances benign and attack classes

TABLE 21. Method-specific controls used in the confirmation matrix.

Method	Adaptation unit	Historical protection	Distinguishing control
DASP	Attributed modules plus classifier	Class-balanced 512-example replay	MMD/conflict router, top-$k = 3$, strict optimizer exclusion
SSF	Full predictor update	Strategic sample selection and forgetting protection	Fixed sample mask, distillation/forgetting control
Experience Replay	Full predictor update	Replay memory	Historical examples mixed with current labels
RepShield	Representation-protection path	Protected representation reference	Robust representation preservation
A-GEM	Full predictor update	Reference gradient memory	Gradient projection against memory direction
Online MLP	Full predictor update	None beyond current state	Direct online supervised update
Online EWC	Full predictor update	Importance-weighted regularization	EWC penalty on important parameters
CDDA-MD	Detector-specific update	Distributional reference	MMD-style drift detector with compact update

implementation, no dropout, batch size 128, AdamW learning rate 10^{-3}, weight decay 10^{-4}, and two triggered/background repair epochs. The router samples 128 module representations, uses four calibration windows, kernel scales $\{0.5, 1, 2\}$, severity threshold 0.15, persistence one, cooldown zero, and top-$k = 3$. The emergency-severity value 0.60 in the locked experimental configuration is inactive in the evaluated primary procedure. Each repair constructs a class-balanced optimization subset of at most 128 new examples from the labeled repair pool, evaluates a disjoint 25% labeled performance holdout, and samples 512 replay examples. Background active modules use plasticity 0.25; emergency modules use $\min\{1, \max(0.25, c_j)\}$ for three epochs. The classifier multiplier is 1 on the base learning rate. Reference refresh is enabled after every applied primary DASP repair.

These values should be regarded as part of the method definition for the reported result. Changing the memory size, router budget, or top-k changes the method and requires a new confirmation experiment. In particular, a full-model update should not be described as DASP merely because the update began with a module-wise score.

APPENDIX D DATASET AND SCENARIO DESIGN

The following design summary explains what is varied and what is held fixed in the prepared streams. A scenario is a transformation of an ordered stream, not a random label permutation. The transformation is applied before model training and is identical for all methods within a paired cell.

The scenario labels are used in the analysis as scientific conditions, not as method-specific hints. DASP does not receive the string ''S7-Local-Low'' as an input. It receives only the stream observations and delayed labels. The scenario name is available to the experiment manager for grouping after the run completes.

A. PROTOCOL VALIDATION

Before a run is accepted, the evaluation procedure verifies that the dataset, scenario, stream seed, and feature count agree with the requested experimental cell. It also verifies that warm-up windows precede the first adaptation window. A mismatch produces an invalid run rather than a silently relabeled result. This is particularly important in a 4,000-run study, where an indexing error could otherwise appear as a statistical effect.

B. CLASS BALANCE AND LABEL ACCOUNTING

The memory sampler maintains class balance whenever both classes are available, and pseudo-labels do not substitute for delayed ground truth. On the primary DASP path, class balancing inspects repair-pool labels before selecting at most 128 new optimization examples, and a disjoint 25% labeled performance holdout is evaluated separately. The recorded 128 field is therefore not a total-label-exposure count. Replay ex-

TABLE 22. Dataset characteristics and intended stress properties.

Dataset	Stream role	Main stress property
CIC-IDS2017	Enterprise benchmark	Multiple attack families and benign variability
NF-BoT-IoT-v3	Botnet/DoS NetFlow benchmark	Localized botnet and denial-of-service changes
NF-CSE-CIC-IDS2018-v3	Enterprise/IoT NetFlow variant	Heterogeneity with high aggregate separability
NF-ToN-IoT-v3	Telemetry-rich IoT benchmark	Difficult evolving traffic composition
NF-UNSW-NB15-v3	Heterogeneous modern-flow benchmark	Ceiling performance and small residual gaps

TABLE 23. Scenario design with temporal behavior, expected failure mode, and DASP diagnostic.

Scenario	Temporal behavior	Expected challenge	Diagnostic emphasis
S0	No deliberate regime change	Unnecessary plasticity	Background-mode stability
S1-Sudden	Abrupt transition	Detection latency	Immediate attribution
S2-Gradual	Distributed transition	Ambiguous attribution	Score persistence and worst cell
S3-Incremental	Several smaller changes	Repeated repairs	Replay and cumulative forgetting
S4-Recurrent	Return of a prior regime	Overwriting and recovery	Replay-conflict behavior
S5-Blip	Brief transient regime	Overreaction	Emergency/background mode behavior
S6-Combined	Compound transformation	Interacting drift signals	Module-count budget
S7-Local-Low	Weak local change	Low signal-to-noise ratio	Positive sparse repair
S7-Local-Medium	Moderate local change	Boundary movement	Attribution precision
S7-Local-High	Strong local change	Multiple-module activation	Top-k limitation

amples remain historical memory rather than new supervision. A strict supervision-matched confirmation requires a rerun in which every method follows one common label-access and holdout contract.

APPENDIX E PREDICTION-FIRST STATE MACHINE

The prediction-first contract is indexed by the outer iteration t and delayed repair index $\tau = t-1$. The model first predicts W_t, computes representation evidence $\mathcal{E}_t$, and enqueues the current item. It then receives Y_τ, restores the queued evidence $\mathcal{E}_\tau$ associated with the same delayed window, computes module-wise gradient conflict, and selects triggered attributed repair, performance-gated emergency repair, or background local consolidation. This textual state machine replaces the older auxiliary diagram so that Figure 1 remains the single canonical algorithm figure.

A. STATE TRANSITION CONDITIONS

The current-window prediction and evidence computation are unconditional and occur before access to Y_t. The delayed transition is controlled by the fixed one-window queue. Triggered attributed repair is controlled by the stored MMD decision D_τ; within that path, conflict attenuates the stored requirement. When D_τ does not trigger, delayed-label performance-holdout Macro-F1 selects emergency or background local repair.

After the routed update, the repair-training subset is added to bounded memory, the enabled reference state is refreshed from the applied model and W_τ, and the routing record is stored. The evaluated DASP procedure does not implement post-update candidate rejection or rollback.

B. EXPANDED PSEUDOCODE

Initialize f_Θ, $\mathcal{H}^{\mathrm{ref}}$, memory $\mathcal{M}$, queue $\mathcal{Q}$, and audit log.
for outer iteration $t = 1, \ldots, T$ with current window W_t **do**
  $\hat{Y}_t \leftarrow f_{\Theta_t}(X_t)$; emit predictions before label access.
  Compute $\mathcal{E}_t = (\mathbf{r}_t, s_t, D_t)$ using the current window and reference snapshot.
  Push $(W_t, \hat{Y}_t, \mathcal{E}_t)$ into $\mathcal{Q}$.
  if labels are available for $\tau = t-1$ **then**
    Pop $(W_\tau, \hat{Y}_\tau, \mathcal{E}_\tau)$ and restore $\mathbf{r}_\tau, s_\tau, D_\tau$.
    Split the delayed window into repair pool and disjoint performance holdout.
    Use repair-pool labels to construct class-balanced C_τ^{new} with at most 128 training examples; draw B_τ^{mem}.
    Compute $c_{j,\tau}$ from C_τ^{new} and B_τ^{mem} gradients.
    if stored MMD decision D_τ triggers **then**
      Set $p_{j,\tau} = r_{j,\tau}/(1 + c_{j,\tau})$ and route positive top-k modules.
    else if $F_\tau^{\mathrm{hold}} < 0.80$ **then**
      Route top-k conflicts for performance-gated emergency repair.
    else
      Route top-k conflicts for background local consolidation.
    end if
    Construct $\widetilde{C}_\tau = C_\tau^{\mathrm{new}} \cup B_\tau^{\mathrm{mem}}$ with new-sample weight 3.
    Exclude all $\Theta_j, j \notin \mathcal{A}_\tau$, from AdamW; train active modules and classifier.
    Update memory and audit; refresh the enabled reference state using W_τ.
    Store the updated model for subsequent windows; no candidate rollback is executed.
  end if
end for

The pseudocode makes two implementation details explicit. First, delayed-label access occurs after prediction, not merely after feature extraction; the 128-example value bounds the

optimization subset rather than total observed labels. Second, the optimizer parameter list changes per repair. A post-hoc mask applied to gradients would not by itself guarantee protection from optimizer state or weight decay.

APPENDIX F STATISTICAL AUDIT OF PAIRED RESULTS

The primary comparison contains 500 paired DASP–SSF cells. The distribution of paired Macro-F1 gaps is asymmetric but centered above zero. The minimum gap is approximately -0.0900, the first quartile is approximately $+0.0002$, the median is $+0.0032$, the third quartile is approximately $+0.0131$, and the maximum is approximately $+0.0684$. The quartiles show that the positive mean is not caused only by a single extreme win, while the negative minimum confirms that individual failures remain operationally relevant.

The gradient-evaluation distribution is especially interpretable. SSF uses 512 evaluations in each paired cell under the locked protocol, whereas DASP is usually at 320 and occasionally increases to 410 when the repair path requires additional work. The cost reduction is therefore not an artifact of a few unusually fast cells. It follows from the module mask and the bounded repair schedule.

A. MULTIPLE-COMPARISON DISCIPLINE

The sign test, paired t-test, and Wilcoxon test are used as complementary analyses of the same primary paired comparison. They should not be presented as three independent discoveries. The sign test is the most robust headline test because it uses only the direction of each paired difference. The other tests provide sensitivity to magnitude and rank structure. Exploratory dataset- and scenario-level comparisons are descriptive and should be interpreted with Holm correction if formal claims are made across many groups.

B. WHAT IS NOT TREATED AS INDEPENDENT

The 16,000 window-level repair records are not counted as 16,000 independent hypothesis tests. They are used to verify structural invariants and diagnose routing behavior. Likewise, the 4,000 runs are not treated as 4,000 independent samples for the DASP–SSF superiority claim; the paired claim uses the 500 cells containing both methods. This distinction prevents pseudo-replication and keeps the inferential unit aligned with the experimental design.

APPENDIX G PERFORMANCE AND MEMORY AUDIT

The performance results have three distinct layers. Prediction throughput measures the rate at which a trained model can score flows. Adaptation time measures the cost of processing delayed labels and updating the state. Memory and resident-set measurements describe the operating footprint. A method can be favorable on one layer and unfavorable on another; therefore no single scalar efficiency score is reported.

DASP's throughput penalty is an architectural consequence of computing four module branches and a fusion classifier. It should not be described as a memory penalty: the model parameter footprint and memory payload are comparable to the other methods in the locked implementation. The practical implication is that a deployment must choose whether the critical resource is inference latency or adaptation latency. A future optimized implementation could fuse the four branches for prediction while retaining separate parameter groups for repair, or could use sparse module execution when the router predicts that some groups are inactive for a known traffic class.

A. UPDATE-TIME INTERPRETATION

The mean update time includes all update intervals. This includes discrepancy calculation, replay sampling, localized training, performance-holdout evaluation on the no-trigger path, and any emergency path. Reporting the mean alone can hide the effect of mode-dependent work, so the median and maximum are included in Table 25. The maximum DASP update time is larger than its mean because a small number of difficult cells invoke extra work. The same pattern is present for SSF, but its entire distribution is shifted upward under the locked schedule.

B. MEMORY INTERPRETATION

The replay capacity is 1,024 records, while a DASP repair uses 512 records. The memory payload reported in the summary is therefore bounded and does not grow with stream length. Peak RSS is affected by the runtime, tensor allocations, and input-data loading, so it is interpreted as an engineering diagnostic rather than a pure model-size measure. The manuscript reports it to make the resource trade-off visible, not to claim a universal memory advantage.

APPENDIX H REPRODUCTION AND ARCHIVAL GUIDE

A complete reproduction requires four components: chronologically prepared streams, the locked experimental configuration, executable method implementations, and row-level plus aggregate analysis records. The archival release should provide these components together with instructions for reproducing the confirmation matrix, statistical comparisons, tables, and figures.

A. RECOMMENDED EXECUTION ORDER

The archival sequence should begin with a small-scale evaluation on one dataset, one scenario, one stream seed, one model seed, and all methods. This initial evaluation verifies software availability, configuration resolution, prediction ordering, run-record structure, and basic metric calculation. The full pilot then covers five datasets, ten scenarios, three stream seeds, one model seed, and eight methods. Only after the full pilot is complete should the confirmation be executed with five stream seeds and two model seeds. The confirmation records are then frozen before manuscript tables are regenerated.

B. FAILURE HANDLING

A failed run should be marked invalid and repeated from the same clean seed rather than silently replaced by a partial summary. If a machine-level failure occurs, the evaluation procedure should retain a diagnostic record and the resolved

TABLE 24. Distribution audit for the 500 paired DASP–SSF Macro-F1 gaps.

Statistic	Minimum	Q1	Median	Q3	Maximum	Mean
Macro-F1 gap	-0.0900	0.0002	0.0032	0.0131	0.0684	0.0078

TABLE 25. Paired adaptation-cost distribution across the same 500 cells.

Metric and method	Minimum	Q1	Median	Q3	Maximum	Mean
DASP update seconds	1.248	1.319	1.357	1.404	5.750	1.415
SSF update seconds	2.428	2.514	2.591	2.680	9.229	2.683
DASP gradient evaluations	320	320	320	320	410	328.7
SSF gradient evaluations	512	512	512	512	512	512.0

TABLE 26. Confirmation-level performance summary by method.

Method	Macro-F1	Update s	Throughput	Grad. eval.	Memory payload	Peak RSS increment
DASP	0.9269	1.415	159,219	328.7	320 KB	28.6 MB
SSF	0.9191	2.683	455,279	512.0	320 KB	28.9 MB
Experience Replay	0.9036	0.546	461,493	192.0	320 KB	28.4 MB
RepShield	0.8881	0.946	464,789	128.0	320 KB	30.1 MB
A-GEM	0.8821	0.495	464,700	128.0	320 KB	29.1 MB
Online MLP	0.8751	0.222	464,101	64.0	320 KB	31.1 MB
Online EWC	0.8751	0.396	454,905	64.0	320 KB	28.9 MB
CDDA-MD	0.7973	0.038	458,585	2.7	320 KB	27.9 MB

TABLE 27. Archival artifact map.

Artifact	Function	Validation use
Chronologically prepared streams	Ordered windows and labels	Verify chronology and feature count
Stream metadata records	Seed and scenario metadata	Verify pairing and experimental-cell identity
Locked experimental configuration	Fixed hyperparameters	Verify method contract
Run summaries	One row per method/cell	Recompute aggregates
Paired cell-level comparison	DASP–SSF join	Recompute tests and quantiles
Protocol-level confirmation record	Run-count and invariant audit	Verify 4,000-run matrix
Generated figures	Visual result summaries	Verify plotted aggregates
LaTeX source	Manuscript and equations	Recompile article

configuration. A result should not enter the paired analysis unless both DASP and SSF completed the same cell. This rule is essential for preserving the meaning of the win count.

C. REPORTING CHECKLIST

Before submission, the authors should replace metadata placeholders, verify all dataset licenses, archive the exact software environment, record the CPU model, and confirm that the final PDF uses the same finalized result records as the source tables. The manuscript should also retain the negative cell and the throughput limitation. Removing either would make the narrative less credible, even if the page count became shorter.

APPENDIX I EXTENDED LIMITATIONS AND FUTURE EXPERIMENTS

The confirmation establishes a useful but bounded result. The streams are controlled chronological artifacts rather than a live multi-site deployment. The feature partition is fixed and contiguous, so the modules are operational subspaces rather than guaranteed semantic concepts. The binary label space does not test retention of individual attack families. The model seeds and stream seeds provide replication but do not exhaust preprocessing, architecture, or hardware variability. These limitations define the next experiments rather than being reasons to reinterpret the current result.

The causal router ablation is particularly important for a journal extension. The current evidence shows that the implementation is structurally local and that the final method performs well. It does not isolate the contribution of the representation discrepancy, supervised conflict, or top-k mask separately. A causal ablation would preserve the same replay, performance-gate, top-k, label, and optimization budgets while comparing random top-k, fixed-module, MMD-only, conflict-only, and combined attribution routing. An all-module variant should be reported separately as a locality ablation because it removes the top-k structural constraint. Such an experiment would turn the attribution narrative from a plausible mechanism into a directly tested causal claim.

The natural-stream experiment is equally important for deployment. Controlled scenarios provide clean diagnostic contrast, but operators encounter changes in service mix, routing policy, benign workload, and attack tooling that do

TABLE 28. Prioritized future experiment matrix.

Experiment	Manipulation	Primary outcome	Priority
Adaptive k	$k \in \{1, 2, 3, 4\}$ by uncertainty	Accuracy versus active-module fraction	High
Learned grouping	Replace contiguous groups with learned groups	Attribution stability	High
Causal router ablation	Shuffle or mask router scores	Accuracy loss and locality	High
Natural streams	Replace controlled scenarios with chronological sites	External validity	High
Multi-class retention	Attack-family labels and open-set attacks	Class-wise forgetting	High
Sparse inference	Skip inactive prediction branches	Throughput recovery	Medium
Label noise	Corrupt delayed labels at several rates	Performance-gate robustness	Medium
Hardware scaling	CPU/GPU and batch-size sweep	Systems portability	Medium

not fit a single synthetic taxonomy. A natural-stream study should keep the prediction-first and delayed-label rules, report site-level pairing, and treat collection site as a higher-level replication factor. It should not simply concatenate all data and call the result "realistic."

Finally, the throughput limitation should be treated as an engineering target. The current architecture executes all branches to obtain attribution-ready representations. A production version could maintain a full-resolution path during adaptation and a fused inference path between repairs. The correctness condition would be that fusion does not change the logits used for the reported predictions, or that any approximation is measured as a separate deployment variant.

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